



“如果觉得有用，请我喝杯茶吧。
咖啡也行。”

甘峰 的赞赏码

分析化学

Analytical Chemistry

甘峰



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Preface

Before retirement, I wrote two analytical chemistry textbooks: *Fundamentals of Analytical Chemistry* (ISBN: 978-7-5025-9676-7) and *Analytical Chemistry* (ISBN: 978-7-03-063519-8). When writing the former, I lacked experience; when writing the latter, I lacked time. The first was a case of not knowing how to write, and the second a case of not having time to write. Taken together, the two barely qualify as having been written.

Yet a traditional printed textbook is frozen the moment it goes to press. Even when the author discovers errors or hits upon a better way of explaining something, he can only wait for the next revision, and that cycle often stretches over years or even more than a decade. Writing a book is, in the end, an exercise in regret.

After retirement, my time is largely my own, and I have naturally lost my best excuse, that of being too busy. So an idea took hold: why not write, at a comfortable pace, a textbook that truly satisfies me? That is how this book came about.

This book is released in electronic form precisely to break free from the limitations of print. It can be revised, supplemented, and refined at any time, so the reader always receives the latest version. In this sense, it is not a finished book but a book that keeps growing; or rather, a book that can never be fully finished.

Writing a book is no easy task; it weighs on the mind and consumes no small amount of coffee. If you find this book helpful, especially if it spares you from failing a course, consider buying me a coffee ! Your support is what keeps me going.

This book is not without its shortcomings. For any errors or infelicities in the text, I would sincerely appreciate your corrections. My usual email addresses are:

cesgf@mail.sysu.edu.cn and ganfeng1964@163.com. I usually reply as promptly as I can; if I do not, I am most likely off somewhere pretending to survey historic sites, as the cover suggests.

GAN, Feng

Guangzhou, 2026 年 9 月 10 日

Chapter 2 Overview of Chemical Quantitative Analysis

Chapter Navigation

We begin with two questions: **What methods are available for chemical quantitative analysis?** (Section 1) and **Are these methods still useful in contemporary science?** (Section 2). Having settled the answers, we step back to build a computational language from the most basic quantities and units (Section 3), and then arm ourselves with four stoichiometric balance relationships (Section 4). Once these four relationships are mastered, every titration computation is simply a matter of selecting one of them to use. Next, we systematically review the core concepts of titration analysis, namely standard solutions, the four titration modes, and indicators (Section 5). Finally, we enter the actual laboratory workflow and learn the complete chain from sampling to reporting of results (Section 6), so as to prepare for the specific titration methods treated in the chapters that follow.

Chemical quantitative analysis is the collective term for the various quantitative analytical methods established upon the chemical reactions between substances and their stoichiometric relationships. In the material world we inhabit, chemical reactions are everywhere; raising such phenomena to a precisely quantifiable level, and thereby achieving accurate determination of the composition of matter, marks a leap forward in human scientific cognition. At the atomic and molecular scale, chemical reactions follow definite stoichiometric laws, and this intrinsic quantitative relationship constitutes the theoretical foundation of analyt-

ical determination, enabling us to infer precisely the content of the analyte from the amount of reactant consumed or the amount of product formed. This chapter provides an overview of the fundamental principles, the classification of methods, and the key content of chemical quantitative analysis in scientific research and practical application.

2.1 Classification of Chemical Quantitative Analysis Methods

This book adopts the now widely accepted classification, dividing chemical quantitative analysis into two broad categories: gravimetric analysis and titrimetric analysis. Gravimetric analysis takes the mass of a substance as the measurement signal; it carries out the determination by weighing the analyte or the mass of its conversion product, and is the most fundamental and direct quantitative analysis method. Titrimetric analysis, by contrast, takes the volume of a standard solution consumed in a chemical reaction as the measurement signal, and completes the determination by precisely measuring the volume consumed in the complete reaction with the analyte; it belongs to a class of indirect quantitative analysis methods that realize quantification in a dynamic process.

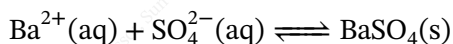
Although gravimetric analysis ultimately weighs a physical mass, its entire process, such as precipitation and volatilization, always depends on specific chemical reactions. This gives it an essential difference from purely physical separation methods, and from it there derive a variety of quantitative analysis techniques with distinct chemical characteristics, fully reflecting the central position of chemical processes in quantitative analysis.

2.1.1 Gravimetric Analysis

Gravimetric analysis is the collective term for a class of quantitative analysis methods that use the weighing technique of the balance for measurement, and it includes precipitation gravimetry, electrogravimetry, volatilization gravimetry, particulate gravimetry, and so on.

Precipitation gravimetry converts the analyte into a precipitate through a chemical reaction, and after filtration, washing, drying or ignition, weighs the precipitate, then calculates the content of the analyte from the chemical composition of the precipitate. Electrogravimetry relies on an electrolytic process to deposit the analyte on an electrode, and determines the content of the analyte by weighing the change in the mass of the electrode before and after the reaction. Volatilization gravimetry heats or otherwise treats the sample so that the volatile constituents escape as a gas, and calculates the content of that constituent from the decrease in the mass of the sample. Particulate gravimetry chiefly separates the particulates from the matrix by physical or chemical means, and accomplishes the quantitative analysis by collecting and weighing them.

This book introduces only precipitation gravimetry and does not further discuss the other types of gravimetric analysis. An example of precipitation gravimetry is:



The characteristic of this reaction is that the barium ion and the sulfate ion react in a 1 : 1 ratio, and the resulting BaSO_4 precipitate has a very small solubility. Once the precipitate is separated out, the content of the barium ion (or sulfate ion) in the sample can be calculated simply by weighing it.

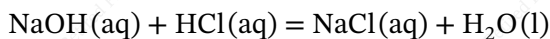
It should be emphasized that although gravimetric analysis is no longer the mainstream method of modern quantitative analysis, it is an absolute method and

constitutes the fundamental basis of chemical quantitative analysis. Any quantitative analysis method must, in the end, trace its quantity value in some manner to the act of weighing on a balance before it can become a trusted method. Although the modern kilogram standard has been defined via the Planck constant since 2019, the dissemination of mass values in the laboratory is still accomplished by weighing on a balance; the relevant content is presented in Section 3 of this chapter.

2.1.2 Titrimetric Analysis

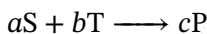
When the analyte cannot be conveniently separated into a solid phase, it can still be made to react quantitatively with another reagent in solution; in that case what must be measured precisely is no longer mass but volume. Titrimetric analysis is based on precisely this idea: it accomplishes quantitative analysis by measuring the volume consumed after the quantitative reaction between substances is complete. For this reason titrimetric analysis was also historically called volumetric analysis. According to the different types of chemical reactions between substances, titrimetric analysis is further divided into acid-base titration, complexation titration, redox titration, and precipitation titration.

For example, the reaction of HCl with NaOH is a typical acid-base reaction, and its reaction equation is as follows:



The characteristic of this reaction is that hydrochloric acid and sodium hydroxide react in a 1 : 1 stoichiometric relationship. Therefore, from the amount of hydrochloric acid consumed one can calculate the amount of sodium hydroxide, and conversely. For a general chemical reaction, we can express it with the

following equation:



If S is the analyte, we may take a certain volume of its solution and react it with a standard solution T of known concentration; from the volume of the T solution consumed and the stoichiometric coefficients a and b between them, we can calculate the content of S.

Relative to gravimetric analysis, titrimetric analysis shows greater applicability and flexibility, and hence its scope of application is broader. At the same time, titrimetric analysis can be traced directly to gravimetric analysis by a certain route, and is therefore also an absolute quantitative analysis method. This book devotes more space to titrimetric analysis.

Key Concept

Gravimetric analysis takes mass as the measurement signal, and mass can be traced directly to the international standard; it therefore does not depend on standard reference materials to operate independently, and is an absolute quantitative method and the foundation of chemical quantitative analysis. Titrimetric analysis takes volume as the measurement signal, deducing the content of the analyte from the volume of standard solution consumed. The concentration of the standard solution must be standardized by gravimetric analysis, so the accuracy of the titrimetric method can ultimately be traced back to the mass standard. One may say that titrimetry is gravimetry extended into the solution system.

2.2 Chemical Quantitative Analysis and Modern Instrumental Analysis

Analytical chemistry has developed to this day into a grand system covering classical chemical analysis and modern instrumental analysis. For the begin-

ner, understanding the position of chemical quantitative analysis (gravimetry and titrimetry) in contemporary analytical science and its functional complementarity with instrumental analysis provides important guidance for learning.

2.2.1 Gravimetry and Titrimetry as the Calibration Cornerstone of Instrumental Analysis

Modern instrumental analysis methods, such as atomic absorption spectrometry, inductively coupled plasma mass spectrometry, and X-ray fluorescence spectrometry, although having notable advantages in sensitivity, speed, and simultaneous multi-element determination, are mostly **relative measurement methods**: the relationship between the instrument response signal (absorbance, intensity ratio, peak area, and so on) and the analyte concentration must first be established by calibration with known-concentration standard substances. The methods used to prepare these standard substances and to assign them accurate concentration values are often precisely gravimetric analysis or titrimetric analysis.

This logical chain can be summarized as: **gravimetry** (determining content directly by weighing on a balance) → **titrimetry** (serving as an intermediate bridge through standard solutions) → **instrumental analysis** (serving as an intermediate bridge through calibration curves). In other words, almost all instrumental analysis methods must be traced back through standard substances to gravimetric analysis as the absolute method, so as to endow their analytical results with a trustworthy “traceability chain of quantity.” Therefore, chemical quantitative analysis is by no means an “old and outdated” technique, but the **metrological foundation** of modern analytical chemistry.

Stop and Think

An instrument gives you a number: 0.452. You do not know whether this number is an absorbance, a transmittance, or a concentration. Please list every link that must be passed from “the instrument’s number” to “the trustworthy content value.” At each link, is the method involved gravimetric, titrimetric, or the standard-curve method?

2.2.2 The Irreplaceability of Chemical Quantitative Analysis in Contemporary Analytical Tasks

Although instrumental analysis has become the mainstay of analytical work, chemical quantitative analysis remains irreplaceable in the following respects:

1. **Standard calibration.** The purity verification and standardization of standard reference materials, such as potassium hydrogen phthalate and potassium dichromate, inherently depend on gravimetric or titrimetric methods.
2. **Determination of major constituents with high accuracy.** When the constituent content falls in the range 1% ~ 100% and the required relative error is at the 0.1% ~ 0.3% level, titrimetric analysis is still one of the methods of choice. For example, the arbitration analysis of iron content in iron ore and the determination of the active ingredient content in medicines are mostly carried out by titrimetric analysis.
3. **Emergency analysis when no standard reference material is available.** For novel compounds or rare samples, it is not uncommon that no commercial standard reference material is yet available. In such cases, the gravimetric or titrimetric method, which performs absolute quantification on the basis of chemical stoichiometric relationships, becomes the

only reliable choice.

4. **Teaching and foundational training.** Training in chemical quantitative analysis cultivates in students a deep understanding of stoichiometric relationships, error propagation, and quality control, core competencies that are indispensable in using any advanced instrument.

2.2.3 Bridging Classical and Modern Methods

It is worth pointing out that many of the concepts and methods introduced in this book form the basis for understanding instrumental analysis. For example:

- The preparation and standardization of standard solutions are applied directly to the preparation of calibration curves in instrumental analysis;
- The rounding rules for significant figures and the methods for uncertainty assessment likewise apply to data processing in instrumental analysis;
- Sampling and sample pretreatment techniques, such as acid digestion, fusion, and extraction, are almost indispensable pretreatment steps for all instrumental analysis methods.

Therefore, learning classical chemical quantitative analysis is not merely for mastering a few specific determination methods, but more importantly for building a panoramic understanding of analytical chemistry: understanding the hierarchical structure of the “traceability chain” and mastering the complete constraint system of “error, precision, and uncertainty,” so that in future practice one can make the most reasonable choice from the toolbox of classical and modern methods according to the needs of a specific task.

Key Concept

The core of the traceability chain: gravimetry (direct weighing) → titrimetry (volume measurement, concentration requiring gravimetric standardization) → instrumental analysis (requiring a calibration curve, standard solutions valued by gravimetry/titrimetry). **Why should one trust an analytical result?** Look at whether it can be traced along this chain back to the international prototype of the kilogram (now defined via the Planck constant).

2.3 Common Quantities and Units

In scientific inquiry, precise quantitative description of physical phenomena is the cornerstone for building theories, testing hypotheses, and realizing technological applications. Whether it is force and energy in physics, substance concentration in chemistry, or population dynamics in biology, all depend on the accurate measurement and standard expression of physical quantities. Therefore, understanding the nature of quantities, the significance of dimensions, and the unification of units has become a basic requirement of interdisciplinary scientific practice.

To promote global scientific cooperation and knowledge sharing, the international scientific community has widely adopted a standardized system of measurement, namely the International System of Units (Le Système international d'unités, SI). Based on seven base quantities, and through rigorous definitions and derivations, this system establishes a complete, consistent, and traceable system of units, providing a reliable and unified basis for measurement for research in different fields.

In chemical quantitative analysis, this foundation is particularly important: what is measured directly on the bench is grams and milliliters, yet chemical reac-

tions occur at the atomic and molecular level; to build a precise conversion bridge between the two, one must rely on the SI system of units. Whether the mass of a substance, its volume, or the stoichiometric relationships in a reaction, all must be accurately measured and calculated on this basis. Therefore, understanding and skillfully applying the International System of Units is not only a basic competency for those engaged in analytical chemistry but also the key to achieving consistency in scientific expression and comparability of results. The subsequent content of this book will take this system as the standard and introduce the various physical quantities involved in chemical quantitative analysis and their units. Table 2.1 lists the seven base quantities of the International System of Units and their units.

Table 2.1: SI base quantities and units

Name of quantity	Name of SI base unit	Symbol of unit
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

On the basis of the SI base units a system of derived units has been established, used in different fields and for different needs. A derived unit is a unit expressed in algebraic form in terms of base units. This section briefly introduces the units relevant to chemical quantitative analysis. For detailed content on quantities and units, please consult GB 3100~3102 – 93.

2.3.1 Mass

Mass is one of the SI base quantities. The symbol for mass is m , and its legal base unit is the kilogram, whose symbol is kg . The mass units commonly used in chemical quantitative analysis are the gram (g), milligram (mg), microgram (μg), and nanogram (ng), and the relationships between them are:

$$1 \text{ kg} = 1 \times 10^3 \text{ g} = 1 \times 10^6 \text{ mg} = 1 \times 10^9 \mu\text{g} = 1 \times 10^{12} \text{ ng} \quad (2.1)$$

The international prototype of the kilogram was established as the standard of mass at the General Conference on Weights and Measures in 1889. It is an alloy artifact of platinum (90%) and iridium (10%) with both diameter and height of 39.17 millimeters, as shown in Figure 2.1. In addition, six official copies are kept together with it at the headquarters of the International Bureau of Weights and Measures in Paris, France. Further copies prepared from the international prototype of the kilogram were distributed to various countries around the world to serve as their national mass standards.

However, even if the international prototype of the kilogram is kept under extremely precise conditions, its mass still undergoes small changes owing to surface adsorption of contaminants. This inherent uncertainty of a standard based on a physical artifact prompted the scientific community to seek a standard of mass based on a fundamental physical constant.

In the 1970s, Dr. Bryan Peter Kibble of the National Physical Laboratory of the United Kingdom invented the watt balance, whose original design was to improve the realization of the unit of electric current, the ampere. With key breakthroughs in quantum metrology, particularly the maturation of the Josephson effect (realizing a quantum association of voltage with frequency, predicted theoretically by Josephson in 1962) and the quantum Hall effect (providing a quantum



Figure 2.1: The international prototype of the kilogram, kept at the headquarters of the International Bureau of Weights and Measures in Paris, France. Image from the Internet.



Figure 2.2: Dr. Bryan Peter Kibble, 1938-2016. Image from the website of the National Physical Laboratory of the United Kingdom.

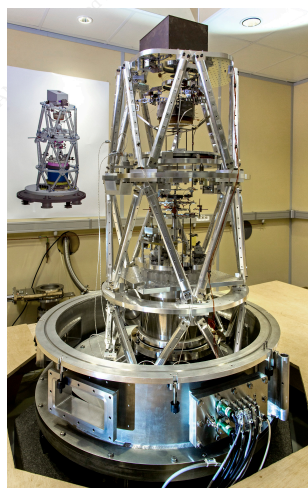


Figure 2.3: The Kibble balance at the International Bureau of Weights and Measures. Image from the BIPM website.

standard of resistance, discovered by Klaus von Klitzing in 1980), the capabilities of this device were greatly extended. The classical Hall effect had been discovered by Edwin Herbert Hall in 1879. People came to realize that the watt balance, through the equivalence of electrical and mechanical power, could relate mass directly to Planck's constant, thereby laying a new quantum foundation for the unit of mass. Dr. Kibble died in 2016, and that same year the device was formally renamed the “Kibble balance” in memory of his pioneering contribution, as shown in Figure 2.3. On the basis of this series of revolutionary achievements, the 26th General Conference on Weights and Measures formally resolved on November 16, 2018 to redefine the kilogram on the basis of Planck's constant, and the new standard of the International System of Units has been implemented worldwide since May 20, 2019, World Metrology Day.

2.3.2 Volume

The volume of an object is the measure of the three-dimensional space that the object itself occupies. In the International System of Units, its derived unit is based on length and is defined as the cubic meter, whose SI symbol is m^3 . In chemical quantitative analysis, because the cubic meter is too large, its fraction units are used instead, such as the cubic decimeter (dm^3), cubic centimeter (cm^3), and cubic millimeter (mm^3). Their conversion relationships are:

$$1 \text{ m}^3 = 10^3 \text{ dm}^3 = 10^6 \text{ cm}^3 = 10^9 \text{ mm}^3 \quad (2.2)$$

In order to conveniently measure the volume of liquids and gases, a dedicated unit more closely matched to everyday use is needed. In 1901 the 3rd General Conference on Weights and Measures first defined the liter as “the volume of 1 kilogram of pure water at its maximum density (about 4°C) and at standard atmospheric pressure,” with symbol L (or l). This definition made it approximately

equal to 1 cubic decimeter¹, and, owing to its convenience in experiments, it rapidly became the most commonly used unit of volume in chemistry, pharmacy, and related fields.

However, subsequent precise measurements found that the liter defined on the basis of the kilogram and the density of water actually has a volume of 1.000028 dm^3 . This small difference cannot be neglected in high-precision measurements. To resolve this contradiction and ensure the unity and rigor of the system of units, in 1964 the 12th General Conference on Weights and Measures resolved to use “liter” as the special name of “cubic decimeter,” and recommended that the cubic decimeter should be preferred in measurements of the highest precision. This decision reconciled historical habit with scientific rigor: it retained the “liter” familiar to chemists and at the same time brought it into the modern framework of the International System of Units based on the meter.

Because the lowercase letter l is easily confused with the digit 1 in writing and printing, the General Conference recommends using L as the symbol for the liter. In chemical quantitative analysis, the milliliter is the most commonly used unit, with symbol mL, and its relationship with the liter is:

$$1 \text{ L} = 1000 \text{ mL} \quad (2.3)$$

It must be noted in particular that the equivalence relation $1 \text{ mL} = 1 \text{ cm}^3$ holds exactly, and it provides the foundation and great convenience for volume conversion in the laboratory.

Mass and volume are the macroscopic physical quantities that we measure directly on the bench. Yet the stoichiometry of chemical reactions occurs at the microscopic level of atoms and molecules. To convert macroscopic quantities such as grams and milliliters into numbers of atoms requires an intermediate bridge

¹More precisely, under this definition $1 \text{ L} = 1.000028 \text{ dm}^3$.

connecting the microscopic and the macroscopic. This bridge starts from the relative mass of atoms, passes through the amount of substance and molar mass, and finally establishes the correspondence between macroscopic mass and the number of microscopic particles. The following subsections develop this thread step by step.

2.3.3 Relative Atomic Mass

Relative atomic mass (symbol A_r) is defined as the ratio of the average mass of the atoms of an element to 1/12 of the mass of an atom of the nuclide ^{12}C . The reason the relative atomic mass system was established is that the absolute mass of atoms is extremely small. For example, the mass of one ^{12}C atom is about 1.99265×10^{-26} kg, which is quite inconvenient to write in practical applications.

Denoting the atomic mass of the nuclide by $m(^{12}\text{C})$, its 1/12 corresponds to a quantity called the atomic mass unit, denoted by the symbol u, as follows:

$$1 \text{ u} = \frac{m(^{12}\text{C})}{12} = 1.66053906660(50) \times 10^{-27} \text{ kg} \quad (2.4)$$

Accordingly, the relative atomic mass of an element E is calculated as follows:

$$A_r(\text{E}) = \frac{m_a(\text{E})}{\text{u}} \quad (2.5)$$

where $m_a(\text{E})$ is the average atomic mass of the element E.

In other words, the relative atomic mass of an element is the ratio of its average atomic mass to 1 u. For example, the relative atomic masses of hydrogen and oxygen are respectively:

$$A_r(\text{H}) = \frac{1.6735 \times 10^{-27} \text{ kg}}{1.66053906660 \times 10^{-27} \text{ kg}} \approx 1.0078$$

$$A_r(\text{O}) = \frac{2.6567 \times 10^{-26} \text{ kg}}{1.66053906660 \times 10^{-27} \text{ kg}} \approx 15.999$$

Notes on symbols and units:

1. Representation for a specific element: when A_r refers to a specific element, the functional form should be used, placing the element symbol in parentheses, for example writing $A_r(\text{H})$ or $A_r(\text{O})$; for a substance that exists as molecules, the relative molecular mass M_r should be used instead (see below). The subscript form should not be used. This is the norm agreed upon by both the international standard and the national standard of China.
2. Representation in the general case: when referring to a basic unit in general, the subscript form is permitted, such as $A_{r,B}$.
3. Importance of dimensions and units: all physical quantities should be accompanied by their units in operations and expressions. Although the units in chemical quantitative analysis are relatively fixed, and the units of intermediate steps may be omitted when the context is clear, it is nevertheless strongly recommended to carry units (dimensions) throughout the calculation. This not only helps to verify the dimensional consistency of the calculation but also fundamentally avoids errors caused by unit confusion.

2.3.4 Relative Molecular Mass

Relative molecular mass is defined as the ratio of the average mass of the molecules of a substance to 1/12 of the mass of an atom of the nuclide ^{12}C , with symbol M_r . According to this definition, the relative molecular mass of a substance B can be calculated as follows:

$$M_r(\text{B}) = \frac{m_m(\text{B})}{u} \quad (2.6)$$

where $m_m(\text{B})$ is the average mass of the molecules of the substance B.

Using the formula above, we can calculate the relative molecular mass of the oxygen molecule. It is known that the average mass of one oxygen atom is about 2.6567×10^{-26} kg, so:

$$\begin{aligned} M_r(\text{O}_2) &= \frac{2 \times 2.6567 \times 10^{-26} \text{ kg}}{1.66053906660 \times 10^{-27} \text{ kg}} \\ &= 2 \times A_r(\text{O}) \\ &= 2 \times 15.999 \\ &= 31.998 \end{aligned} \quad (2.7)$$

From the calculation above it is clear that the relative molecular mass of a molecule equals the sum of the relative atomic masses (A_r) of all the atoms composing that molecule. This is the most core and most practical rule of calculation for the relative molecular mass. Therefore, the relative molecular mass of sodium hydroxide can be calculated as follows:

$$\begin{aligned} M_r(\text{NaOH}) &= A_r(\text{Na}) + A_r(\text{O}) + A_r(\text{H}) \\ &= 22.990 + 15.999 + 1.0078 \\ &= 39.997 \end{aligned} \quad (2.8)$$

It should be pointed out in particular that for ionic compounds (such as NaCl) or covalent network crystals (such as SiO_2), which possess no independent molecules in the strict sense, the concept of “relative molecular mass” is not strictly applicable. In such cases a more general term should be adopted, namely relative formula mass or relative mass of the formula unit, whose definition and rules of calculation are the same as those of the relative molecular mass: the ratio of the average mass of one formula unit to u . For example:

$$M_r(\text{NaCl}) = A_r(\text{Na}) + A_r(\text{Cl}) = 22.990 + 35.45 \approx 58.44$$

$$M_r(\text{CaCO}_3) = A_r(\text{Ca}) + A_r(\text{C}) + 3A_r(\text{O}) \approx 100.09$$

In chemical quantitative analysis this concept is crucial, because the calculations involved are typically based on the chemical formula of a substance rather than its molecular composition.

Relative atomic mass and relative molecular mass (or, more generally, relative formula mass) are dimensionless physical quantities; they relate the mass of microscopic particles to a unified atomic standard. Their core value lies in additivity, which allows chemists to conveniently compute the relative mass of any substance of known chemical formula by consulting tables of relative atomic masses and relative molecular masses.

2.3.5 Amount of Substance

Amount of substance is one of the base physical quantities of the International System of Units; it was established specifically for counting the number of basic units and provides the basis for the quantitative description of chemical reactions. In the long history of the development of chemistry, quantitative determination of the reactions of compounds has always been one of the core problems of research. In 1971 the General Conference on Weights and Measures formally established “amount of substance” as a base physical quantity of the SI, defining its unit as the mole (mol), which at the time was premised on its association with the number of atoms in 0.012 kg of carbon-12. With the progress of metrology, in 2018 the 26th General Conference on Weights and Measures adopted a redefinition of the mole, which took effect on May 20, 2019. The new definition states that 1 mole contains exactly $6.02214076 \times 10^{23}$ elementary entities, that is, the mole is defined by fixing the numerical value of the Avogadro constant. This definition freed the mole from its dependence on carbon-12 and removed the coupling

between the definition of the mole and that of the kilogram.

Suppose a system contains a substance S. When the basic unit of S is S itself, if the number of basic units is N_S , then the amount of substance of S can be calculated as follows:

$$n_S = \frac{N_S}{N_A} \quad (2.9)$$

where N_A is the Avogadro constant², whose numerical value is about $6.02214076 \times 10^{23} \text{ mol}^{-1}$.

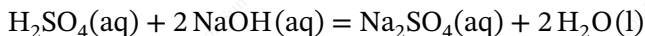
From a popular point of view, the amount of substance is in effect determined by “counting” the number of basic units. That is, if the number of basic units of S equals $6.02214076 \times 10^{23}$, then the amount of substance of S is said to be 1 mol.

The basic unit referred to here can be an atom, a molecule, an ion, an electron, and so on, or a specific combination of these particles. This specific combination may exist objectively, or it may be conceived according to the stoichiometric relationship of a chemical reaction; it may be an integral combination of particles, or a non-integral combination of particles.

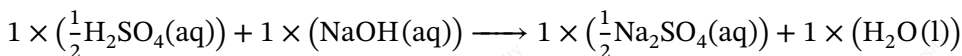
For example, the actual species of sulfuric acid is H_2SO_4 , and the amount of sulfuric acid is usually expressed in terms of H_2SO_4 . In this case, 1 mol of sulfuric acid contains $6.02214076 \times 10^{23}$ H_2SO_4 molecules. However, sometimes the stoichiometric coefficient of the chemical reaction involving sulfuric acid is not 1. In such a case, if the whole is “split,” it is very beneficial for establishing a simple stoichiometric relationship for the entire reaction. For example, if a sulfuric acid molecule is subjected to a “splitting operation,” the basic unit at this point is $\frac{1}{2}\text{H}_2\text{SO}_4$, and the corresponding “number of particles” of this basic unit becomes twice that of the actual species.

²Historically the symbol L was used for the Avogadro constant.

To help the reader understand this concept, let us first start from a very simple example. For the acid-base neutralization reaction:



If the conventional actual species is taken as the basic unit, the stoichiometric relationship between sulfuric acid and sodium hydroxide is $n(\text{NaOH}) = 2n(\text{H}_2\text{SO}_4)$. But if $\frac{1}{2}\text{H}_2\text{SO}_4$ is taken as the basic unit of sulfuric acid and NaOH as the basic unit of sodium hydroxide, the reaction equation can be rewritten as:



At this point the stoichiometric coefficients are all unified to 1, and the stoichiometric relationship is directly simplified to $n(\frac{1}{2}\text{H}_2\text{SO}_4) = n(\text{NaOH})$. It must be emphasized in particular that: **the basic unit can be arbitrarily constructed, but the mass of the substance does not change when the basic unit is changed.** For example, 0.1 mol of H_2SO_4 (mass 9.8 g) corresponds to 0.2 mol of $\frac{1}{2}\text{H}_2\text{SO}_4$, whose mass is still 9.8 g.

Stop and Think

If the reaction is $\text{H}_3\text{PO}_4 + 3\text{NaOH}$, how would you construct the basic units so that all the stoichiometric coefficients become 1?

The following example illustrates the application of this construction method in actual analysis.

Example 2.1 A nitrogen-fertilizer sample is a mixture of ammonium sulfate and ammonium nitrate. A mass m g of the sample is weighed and added to an excess of concentrated sodium hydroxide solution and heated, so that all the ammonium ions of the two ammonium salts are converted

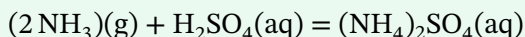
into ammonia and driven off. The evolved ammonia is passed into a sulfuric acid solution of volume $V(\text{H}_2\text{SO}_4)$ L and concentration $c(\text{H}_2\text{SO}_4)$ mol/L. After the reaction is complete, the remaining sulfuric acid is just completely neutralized by $V(\text{NaOH})$ L of a sodium hydroxide solution of concentration $c(\text{NaOH})$ mol/L. Calculate the amounts of substance of ammonium sulfate and ammonium nitrate in the nitrogen-fertilizer sample.

Solution: Let the amount of substance of ammonium sulfate in the sample be $n((\text{NH}_4)_2\text{SO}_4)$ and that of ammonium nitrate be $n(\text{NH}_4\text{NO}_3)$. Since 1 mol of ammonium sulfate contains 2 mol of ammonium ions, the number of moles of ammonia that the sample can produce is:

$$n(\text{NH}_3) = 2n((\text{NH}_4)_2\text{SO}_4) + n(\text{NH}_4\text{NO}_3)$$

The relation above uses the facts that each mole of ammonium sulfate contains 2 moles of ammonium groups and each mole of ammonium nitrate contains 1 mole of ammonium group.

The ammonia produced is absorbed into the sulfuric acid, and the chemical reaction equation is:



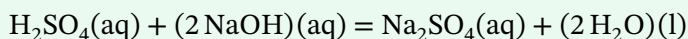
Let the amount of sulfuric acid consumed be $n_1(\text{H}_2\text{SO}_4)$, then the stoichiometric relationship is:

$$n(2\text{NH}_3) = n_1(\text{H}_2\text{SO}_4)$$

That is:

$$\frac{1}{2}n(\text{NH}_3) = n_1(\text{H}_2\text{SO}_4)$$

The remaining sulfuric acid is neutralized by sodium hydroxide, and the neutralization reaction is:



The stoichiometric relationship is:

$$n_2(\text{H}_2\text{SO}_4) = n(2\text{NaOH})$$

That is:

$$n_2(\text{H}_2\text{SO}_4) = \frac{1}{2}n(\text{NaOH})$$

The relation above uses the fact that 2NaOH reacts with H_2SO_4 in a 1 : 1 stoichiometric ratio.

Therefore:

$$\begin{aligned} c(\text{H}_2\text{SO}_4)V(\text{H}_2\text{SO}_4) &= n_1(\text{H}_2\text{SO}_4) + n_2(\text{H}_2\text{SO}_4) \\ &= n((\text{NH}_4)_2\text{SO}_4) + \frac{1}{2}n(\text{NH}_4\text{NO}_3) + \frac{1}{2}n(\text{NaOH}) \end{aligned}$$

The expression above shows that the total amount of sulfuric acid consumed equals the sum of the amount consumed in absorbing the ammonia and the amount of the remaining sulfuric acid neutralized by NaOH .

Therefore:

$$n((\text{NH}_4)_2\text{SO}_4) + \frac{1}{2}n(\text{NH}_4\text{NO}_3) = c(\text{H}_2\text{SO}_4)V(\text{H}_2\text{SO}_4) - \frac{1}{2}n(\text{NaOH})$$

Using the relationship between the amount of substance and the mass of the substance, we obtain:

$$n((\text{NH}_4)_2\text{SO}_4)M((\text{NH}_4)_2\text{SO}_4) + n(\text{NH}_4\text{NO}_3)M(\text{NH}_4\text{NO}_3) = m$$

Solving these two equations yields the amounts of substance of ammonium sulfate and ammonium nitrate. (Further details are omitted.)

Common Pitfall

After changing the basic unit, the mass of the substance does not change, but the concentration does. For example, when $c(\text{H}_2\text{SO}_4) = 0.10 \text{ mol/L}$, $c(\frac{1}{2}\text{H}_2\text{SO}_4) = 0.20 \text{ mol/L}$; this is a doubling, not a halving. Many beginners get this ratio backwards.

The introduction of the amount of substance n provides chemistry with a standard unit for counting collections of microscopic particles, namely $1 \text{ mol} = N_{\text{A}}$ entities. However, on the bench, we cannot directly count microscopic particles; we can only weigh their mass with a balance. This raises a core problem: how many grams of a substance should be weighed out so that it corresponds exactly to 1 mol of that substance? To resolve the conversion need between mass and amount of substance, the concept of molar mass is introduced.

2.3.6 Molar Mass

Molar mass is a core physical quantity in chemical quantitative analysis. It is defined as the ratio of the mass m of a substance to its amount of substance n , namely:

$$M = \frac{m}{n} \quad (2.10)$$

The SI unit of molar mass is the kilogram per mole (kg/mol). However, in chemical quantitative analysis, the numerical value expressed in this unit is too small and is quite inconvenient in practice. Therefore, the chemical field commonly adopts its decimal fraction unit, namely the gram per mole (g/mol).

Since 1 mol contains $6.02214076 \times 10^{23}$ basic units, the molar mass can be calculated as follows:

$$M(\text{S}) = m(\text{S}) \times N_{\text{A}} \quad (2.11)$$

where $m(S)$ is the mass of the basic unit S .

For example, the mass of one oxygen atom is 2.6567×10^{-26} kg, so the molar mass of oxygen is:

$$\begin{aligned} M(\text{O}) &= 2.6567 \times 10^{-26} \text{ kg} \times 6.02214076 \times 10^{23} \text{ mol}^{-1} \\ &= 1.600 \times 10^{-2} \text{ kg/mol} \\ &= 16.00 \text{ g/mol} \end{aligned} \quad (2.12)$$

In this example it can be seen that, when the molar mass is expressed in g/mol, the molar mass of the oxygen atom is numerically equal to its relative atomic mass.

Stop and Think

Is this a coincidence? Why must the numerical value of the molar mass under the unit g/mol equal the relative atomic mass? Please try starting from the definitions of M and M_r to see whether you can find the intrinsic connection between the two.

The derivation below proves that this is a general rule. For a substance S , we divide it into three steps to examine it.

First, we start from the definition of molar mass. According to Equation 2.11, $M(S) = m(S) \times N_A$, which means that the mass of 1 mol of the substance equals the mass of a single basic particle multiplied by the Avogadro constant. And according to the definition of relative atomic (molecular) mass, the mass of a single particle can be written as the relative mass multiplied by the atomic mass unit: $m(S) = M_r(S) \times u$. Substituting gives:

$$M(S) = M_r(S) \times u \times N_A$$

Next is the key step: substituting the definition related to ^{12}C . According to the definition of the atomic mass unit u , $u = m(^{12}\text{C})/12$, and the mass of

1 mol of ^{12}C atoms is 0.012 kg^3 , so the mass of a single ^{12}C atom is $m(^{12}\text{C}) = 0.012\text{ kg/mol} / N_{\text{A}}$. Substituting these two relationships in turn:

$$M(\text{S}) = M_{\text{r}}(\text{S}) \times \frac{m(^{12}\text{C})}{12} \times N_{\text{A}} = M_{\text{r}}(\text{S}) \times \frac{0.012\text{ kg/mol}}{N_{\text{A}} \times 12} \times N_{\text{A}}$$

At this point, note that the N_{A} in the numerator and the N_{A} in the denominator cancel exactly:

$$M(\text{S}) = M_{\text{r}}(\text{S}) \times \frac{0.012\text{ kg/mol}}{12}$$

Finally, $0.012/12 = 0.001$, and $0.001\text{ kg/mol} = 1\text{ g/mol}$. Putting the entire derivation together:

$$\begin{aligned} M(\text{S}) &= m(\text{S}) \times N_{\text{A}} && \text{(definition of molar mass)} \\ &= M_{\text{r}}(\text{S}) \times u \times N_{\text{A}} && \text{(definition of relative mass)} \\ &= M_{\text{r}}(\text{S}) \times \frac{m(^{12}\text{C})}{12} \times N_{\text{A}} && \text{(definition of } u\text{)} \\ &= M_{\text{r}}(\text{S}) \times \frac{0.012\text{ kg/mol}/N_{\text{A}}}{12} \times N_{\text{A}} && \text{(molar mass of } ^{12}\text{C}\text{)} \\ &= M_{\text{r}}(\text{S}) \times \frac{0.012\text{ kg/mol}}{12} && (N_{\text{A}} \text{ cancels)} \\ &= M_{\text{r}}(\text{S}) \times 0.001\text{ kg/mol} && \text{(simplify)} \\ &= M_{\text{r}}(\text{S})\text{ g/mol} && \text{(unit conversion)} \end{aligned}$$

The conclusion of the derivation is very concise: when the molar mass is expressed in g/mol, its numerical value always equals the relative atomic (molecular) mass. This is not a coincidence but a natural consequence of the precise definitions of the mole, the kilogram, and the atomic mass unit in the International System of Units.

³Before the redefinition of the mole in 2019 this relationship was precisely the definition of the mole; after the redefinition it no longer has this defining status, yet it still holds with a relative accuracy better than 10^{-9} , so the conclusion of this derivation is unaffected.

With this rule, the molar mass of a molecule can be conveniently calculated from the molar masses of its atoms. For example, the molar mass of HCl is:

$$M(\text{HCl}) = 1.0078 \text{ g/mol} + 35.45 \text{ g/mol} = 36.46 \text{ g/mol} \quad (2.13)$$

Through the molar mass of a substance, the amount of substance (that is, the number of moles of the substance) can be easily calculated. For a substance S, the calculation formula for its amount of substance is as follows:

$$n(\text{S}) = \frac{m(\text{S})}{M(\text{S})} \quad (2.14)$$

where $n(\text{S})$, $m(\text{S})$, and $M(\text{S})$ are the amount of substance, the mass, and the molar mass of the substance S, respectively.

Example 2.2 The concentrated hydrochloric acid solution used in the laboratory contains 37% of HCl, and the density of the solution is 1.19 g/mL. Calculate the amount of substance of HCl in this concentrated hydrochloric acid.

Solution: The amount of substance of HCl contained in 1.0 mL of this concentrated hydrochloric acid solution is:

$$n(\text{HCl}) = \frac{1.0 \text{ mL} \times 1.19 \text{ g/mL} \times 37\%}{36.46 \text{ g/mol}} = 0.012 \text{ mol}$$

2.3.7 Amount-of-Substance Concentration

Suppose there is a substance A whose amount of substance is n_{A} . The amount-of-substance concentration of A is defined as:

$$c_{\text{A}} = \frac{n_{\text{A}}}{V} \quad (2.15)$$

where the lowercase character c_A denotes the amount-of-substance concentration of A, whose SI unit is the mole per cubic meter, and which in chemistry is customarily expressed in mol/L.

From $c = n/V$ a formula frequently used in the laboratory can be derived: when a solution is diluted, the amount of substance of the solute does not change ($n_1 = n_2$), so $c_1 V_1 = c_2 V_2$. For example, diluting 10.00 mL of a 0.1000 mol/L HCl solution to 100.0 mL gives the new concentration $c_2 = 0.1000 \times 10.00/100.0 = 0.01000$ mol/L. This basic relationship will be used frequently in the course of chemical quantitative analysis.

The amount-of-substance concentration of a substance A is also called the molar concentration of A, and is sometimes simply referred to as the concentration of A.

Example 2.3 The concentrated sulfuric acid used in the laboratory contains 96% of H_2SO_4 , and at 25°C its density is 1.840 g/cm³. Calculate the molar concentration of this concentrated sulfuric acid.

Solution: The mass of H_2SO_4 contained in 1 L of this solution is:

$$m(\text{H}_2\text{SO}_4) = 1000 \text{ mL} \times 1.840 \text{ g/mL} \times 96\% = 1.8 \times 10^3 \text{ g}$$

The number of moles of sulfuric acid is:

$$n(\text{H}_2\text{SO}_4) = \frac{1.8 \times 10^3 \text{ g}}{98.08 \text{ g/mol}} = 18 \text{ mol}$$

Therefore, the molar concentration of this sulfuric acid solution is:

$$c(\text{H}_2\text{SO}_4) = \frac{18 \text{ mol}}{1 \text{ L}} = 18 \text{ mol/L}$$

The two examples above show the general method of calculating the molar concentration from a commercial reagent of known mass fraction ω and density ρ :

$$c = \frac{\omega \rho}{M} \quad (2.16)$$

where ω is the mass fraction, ρ is the density of the solution (g/L), and M is the molar mass (g/mol). Note that both ω and M are quantities related to the basic unit, so the calculation must ensure that the two refer to the same basic unit. This formula is the most common bridge connecting the label of a reagent bottle with the titration calculation in the laboratory.

Likewise, the amount-of-substance concentration is also a quantity related to the basic unit, and the basic unit must be specified when it is applied. If the basic unit changes, the numerical value of the amount-of-substance concentration should change accordingly. For example, if $c(\text{H}_2\text{SO}_4) = 0.10 \text{ mol/L}$, then $c\left(\frac{1}{2}\text{H}_2\text{SO}_4\right) = 0.20 \text{ mol/L}$.

It should be noted here that in chemical quantitative analysis the amount-of-substance concentration c_A of A is also called the molar analytical concentration of A, abbreviated as the analytical concentration of A. It denotes the total number of moles of A in 1 liter of solution, so it is sometimes also called the total molar concentration of A. The reason these terms are assigned to the amount-of-substance concentration is mainly to make the original species of the chemical substance traceable.

For example, the original species of H_2SO_4 is H_2SO_4 , but when it is dissolved in water, only HSO_4^- and SO_4^{2-} species exist in the resulting solution, and no H_2SO_4 species is present. To express the concentrations of the various particular species of such substances in the equilibrium state, the term molar equilibrium concentration (or equilibrium concentration) is introduced, and square brack-

ets are correspondingly used to denote the equilibrium concentration. Thus the equilibrium concentration of the hydrogen sulfate ion is $[\text{HSO}_4^-]$, while the equilibrium concentration of the sulfate ion is $[\text{SO}_4^{2-}]$, and their units are still mol/L. The sum of the equilibrium concentrations of these two species equals the molar analytical concentration of sulfuric acid.

Example 2.4 Dissolve 0.2401 g of sodium hydrogen sulfate in 200.0 mL of water, and calculate the equilibrium concentrations of the various species in the solution. What is their relationship with the total concentration? It is known that the degree of dissociation of HSO_4^- is 10.0%.

Solution: The number of moles of sodium hydrogen sulfate is:

$$n(\text{NaHSO}_4) = \frac{0.2401 \text{ g}}{120.06 \text{ g/mol}} = 0.002000 \text{ mol}$$

Since sodium hydrogen sulfate dissociates completely, and the resulting hydrogen sulfate ion in turn undergoes 10.0% dissociation, we have:

$$n(\text{HSO}_4^-) = 0.002000 \text{ mol} \times (100\% - 10.0\%) = 1.800 \times 10^{-3} \text{ mol}$$

$$n(\text{SO}_4^{2-}) = 0.002000 \text{ mol} \times 10.0\% = 2.000 \times 10^{-4} \text{ mol}$$

Therefore:

$$[\text{HSO}_4^-] = \frac{1.800 \times 10^{-3} \text{ mol}}{200.0 \times 10^{-3} \text{ L}} = 9.000 \times 10^{-3} \text{ mol/L}$$

$$[\text{SO}_4^{2-}] = \frac{2.000 \times 10^{-4} \text{ mol}}{200.0 \times 10^{-3} \text{ L}} = 1.000 \times 10^{-3} \text{ mol/L}$$

The total concentration of sodium hydrogen sulfate is:

$$c(\text{NaHSO}_4) = \frac{0.002000 \text{ mol}}{200.0 \times 10^{-3} \text{ L}} = 1.000 \times 10^{-2} \text{ mol/L}$$

The result of the calculation shows:

$$c(\text{NaHSO}_4) = [\text{HSO}_4^-] + [\text{SO}_4^{2-}]$$

Stop and Think

Analytical concentration c and equilibrium concentration $[]$ use different symbols for a reason: the former traces back to the total amount of the original species, while the latter counts only the actual species present in solution. If the two are conflated, all subsequent calculations in acid-base equilibrium and complexation equilibrium will go wrong. Use the following example to test whether you have really distinguished them:

For a NaHSO_4 solution (it is known that HSO_4^- undergoes 10% dissociation into SO_4^{2-} and H^+), judge whether the following equations are correct, and briefly state the reason:

$$(1) c(\text{NaHSO}_4) = [\text{Na}^+]$$

$$(2) c(\text{NaHSO}_4) = [\text{HSO}_4^-]$$

$$(3) [\text{Na}^+] = [\text{HSO}_4^-] + [\text{SO}_4^{2-}]$$

Key Concept

The conversion chain from weighing to concentration: weigh the sample (mass m) $\rightarrow$ divide by the molar mass ($n = m/M$) to obtain the amount of substance $\rightarrow$ divide by the volume of the solution ($c = n/V$) to obtain the concentration. The three physical quantities on this chain, namely mass, amount of substance, and concentration, are the basis of all quantitative calculation. The distinction between analytical concentration c and equilibrium concentration $[]$ is likewise the premise of all subsequent equilibrium calculations: the former traces back to the original species, while the latter counts the actual species.

2.4 Stoichiometric Balance Relationships

Stop and Think

Before learning the content that follows, please confirm that you can answer the following questions:

1. Given the mass m and the molar mass M of a substance, how do you calculate the amount of substance n ?
2. After rewriting the basic unit of H_2SO_4 as $\frac{1}{2}\text{H}_2\text{SO}_4$, how many times does the concentration become? Does the mass change?
3. What is the difference between analytical concentration c and equilibrium concentration $[]$?

The three items above are the cornerstone of all stoichiometric balance calculations. If you are uncertain about any of them, please return to Section 2.3 and study it again.

Chemical quantitative analysis is built upon chemical reactions that involve microscopic species such as atoms, ions, or molecules, and consequently the relationships between the amounts of substances are usually described directly in terms of these species. On the other hand, since the concept of “amount of substance” is itself based on a count of particles, it allows us to flexibly construct a suitable “basic unit” according to the specific need, so as to express the stoichiometric relationships among the components of a reacting system, even when the constructed basic unit is not an existing form of the substance. See Example 2.1.

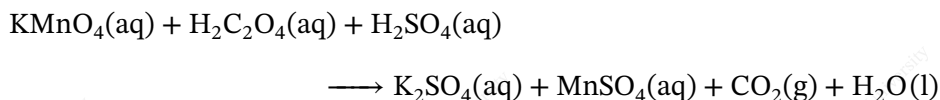
The reason this method is feasible lies in the fundamental principle of conservation of atoms in a chemical reaction: before and after the reaction, the kinds and the numbers of atoms remain unchanged, and they are merely rearranged and transferred from one substance to another. Using this core conservation principle, the various stoichiometric balance relationships in a reacting system can

be established and solved. This chapter generalizes them into four basic types: amount-of-substance balance, elemental balance, charge balance, and conservation of electrons. Once these four balance relationships are mastered, any titration calculation is nothing more than applying one of them. **When faced with a specific problem, make the choice according to the following principles:**

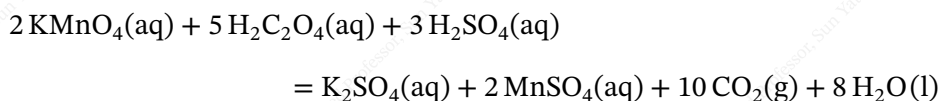
- **Amount-of-substance balance** — applicable to a balanced reaction with definite stoichiometric coefficients, the most direct method;
- **Elemental balance** — used when we do not wish to, or need not, balance the complete reaction equation, directly tracing which atoms belong to the reactants and which to the products;
- **Charge balance** — deals with the coexistence of multiple species (acids, bases, complexes) in solution, constraining the relationships among the concentrations of the ions;
- **Conservation of electrons** — exclusive to redox reactions, using the number of electrons gained or lost by each species as a bridge to establish stoichiometric relationships.

2.4.1 Amount-of-Substance Balance

Amount-of-substance balance describes a stoichiometric relationship between the amount of one substance and the amount of another in a reacting system. For example, potassium permanganate and oxalic acid react in acid solution as follows:



Since the numbers of atoms of all the elements before and after the reaction must be equal, the balanced reaction equation is obtained:



From this the balance relationship between the amount of potassium permanganate and the amount of oxalic acid can be obtained:

$$2 \text{ mol KMnO}_4 \sim 5 \text{ mol H}_2\text{C}_2\text{O}_4 \quad (2.17)$$

Following the approach of Example 2.1, the reaction equation above can also be re-expressed in the following way:

$$\begin{aligned} 1 \times (2\text{KMnO}_4(\text{aq})) + 1 \times (5\text{H}_2\text{C}_2\text{O}_4(\text{aq})) + 1 \times (3\text{H}_2\text{SO}_4(\text{aq})) \\ = 1 \times (\text{K}_2\text{SO}_4(\text{aq})) + 1 \times (2\text{MnSO}_4(\text{aq})) + 1 \times (10\text{CO}_2(\text{g})) + 1 \times (8\text{H}_2\text{O}(\text{l})) \end{aligned}$$

In this mode of expression, different basic units are adopted for different substances, which on the surface appears to increase the complexity. However, it provides a more concise expression of the amount-of-substance balance relationship, because at this point the stoichiometric coefficients among the substances are all 1, and the amount-of-substance balance relationship can be established very conveniently. For example, if we express the amount of potassium permanganate as $n(2 \text{KMnO}_4)$ and the amount of oxalic acid as $n(5 \text{H}_2\text{C}_2\text{O}_4)$, then:

$$1 \times n(2\text{KMnO}_4) = 1 \times n(5\text{H}_2\text{C}_2\text{O}_4) \quad (2.18)$$

The following example demonstrates the advantage of using this mode of expression.

Example 2.5 The oxalic acid content of spinach can be measured by the potassium permanganate method. An extract containing oxalic acid is obtained from spinach with a solvent. A potassium permanganate solution of concentration 0.02800 mol/L is added to the extract, and 12.50 mL of this permanganate solution is consumed before the oxalic acid has completely reacted. Find the mass of oxalic acid in the extract.

Solution: The amount of potassium permanganate consumed is:

$$n(\text{KMnO}_4) = 12.50 \times 10^{-3} \text{ L} \times 0.02800 \text{ mol/L} = 3.500 \times 10^{-4} \text{ mol}$$

Therefore:

$$\begin{aligned} n(2\text{KMnO}_4) &= \frac{1}{2} \times n(\text{KMnO}_4) \\ &= \frac{1}{2} \times 3.500 \times 10^{-4} \text{ mol} \\ &= 1.750 \times 10^{-4} \text{ mol} \end{aligned}$$

Since:

$$n(5\text{H}_2\text{C}_2\text{O}_4) = n(2\text{KMnO}_4)$$

the number of moles of oxalic acid in the extract is:

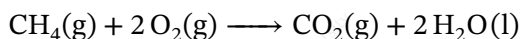
$$\begin{aligned} n(\text{H}_2\text{C}_2\text{O}_4) &= 5 \times n(5\text{H}_2\text{C}_2\text{O}_4) \\ &= 5 \times n(2\text{KMnO}_4) \\ &= 5 \times 1.750 \times 10^{-4} \text{ mol} \\ &= 8.750 \times 10^{-4} \text{ mol} \end{aligned}$$

The mass of oxalic acid in the extract is:

$$m(\text{H}_2\text{C}_2\text{O}_4) = 8.750 \times 10^{-4} \text{ mol} \times 90.04 \text{ g/mol} = 0.07879 \text{ g}$$

2.4.2 Elemental Balance

Elemental balance is the direct embodiment of the law of conservation of mass at the atomic level: a chemical reaction neither destroys nor creates atoms, so the total number of atoms of any element before and after the reaction must be equal. Using this principle, one need only determine the attribution of a certain element to the reactants and the products, and the stoichiometric relationship between the two amounts of substance can be written directly, without relying on a balanced complete reaction equation. For example, for the following combustion reaction:



After complete combustion, the carbon and hydrogen atoms of CH_4 are entirely converted into CO_2 and H_2O . If the amounts of substance of the three are $n(\text{CH}_4)$, $n(\text{CO}_2)$, and $n(\text{H}_2\text{O})$, respectively, then the elemental balance is expressed as follows:

$$1 \times n(\text{CH}_4) = 1 \times n(\text{CO}_2) \quad (2.19)$$

$$4 \times n(\text{CH}_4) = 2 \times n(\text{H}_2\text{O}) \quad (2.20)$$

Although Equation 2.19 and Equation 2.20 can also be obtained by balancing the reaction equation, they are more easily established through the principle that the total number of atoms of an element remains unchanged. One CH_4 molecule contains 1 carbon atom and 4 hydrogen atoms, so $n(\text{CH}_4)$ moles correspond to $1 \times n(\text{CH}_4)$ moles of carbon atoms and $4 \times n(\text{CH}_4)$ moles of hydrogen atoms, which are respectively equal to the total amounts of carbon and hydrogen in the products.

The principle of elemental balance can also be used in the case where an atomic group is taken as the basic unit, and the reason is simple: an

atomic group is itself made up of atoms. For example, each formula unit of $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6 \text{H}_2\text{O}$ contains two NH_4^+ groups, that is, two nitrogen atoms. If NH_4^+ is taken as a basic unit, then, according to the conservation of the element nitrogen, the relationship between the amount of NH_4^+ released upon complete dissociation of this formula unit and the formula unit itself is:

$$2 \times n((\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6 \text{H}_2\text{O}) = 1 \times n(\text{NH}_4^+) \quad (2.21)$$

Example 2.6 To determine the magnesium oxide content of a coating, a sample of the coating is taken and, after dissolution, precipitation, filtration, and drying, 0.4500 g of MgNH_4PO_4 is obtained. Calculate the mass of MgO in the coating.

Solution: According to the elemental balance of the magnesium atom, the following relationship is obtained:

$$1 \times n(\text{MgO}) = 1 \times n(\text{MgNH}_4\text{PO}_4)$$

Therefore the mass of magnesium oxide in the coating is:

$$\begin{aligned} m(\text{MgO}) &= n(\text{MgO}) \times M(\text{MgO}) \\ &= n(\text{MgNH}_4\text{PO}_4) \times M(\text{MgO}) \\ &= \frac{0.4500 \text{ g}}{137.32 \text{ g/mol}} \times 40.30 \text{ g/mol} \\ &= 0.1321 \text{ g} \end{aligned}$$

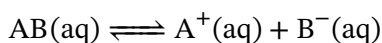
It should be noted that the “material balance” (also called mass balance) accounting method common in the chemical industry is essentially based on the same principle of the conservation of the number of atoms of the elements, the

only difference being that the scale of application is extended from a single reaction to the input-output accounting of an entire process. Therefore, elemental balance is both the basic tool for deriving the stoichiometric equation of a chemical reaction and the logical starting point for understanding material balance at a larger scale.

2.4.3 Charge Balance

Matter in nature is usually an electrically neutral body. Some readily ionizable substances dissociate when dissolved in water, and positive and negative ions then appear in the solution. Because the total positive charge carried by the positive ions equals the total negative charge carried by the negative ions, the whole solution system remains electrically neutral. This characteristic of the solution system becomes the charge balance condition that constrains the concentration relationships among the various species.

Suppose a molecule of the form AB has the following dissociation equilibrium in aqueous solution:



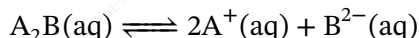
Charge balance requires:

$$\text{number of } \text{A}^+ \text{ particles} = \text{number of } \text{B}^- \text{ particles} \quad (2.22)$$

Note that the reason for expressing this in terms of the species present in the solution is that these are the bodies that carry the positive and negative charges. Since the number of particles is equivalent to the amount of substance, which in turn corresponds to the amount-of-substance concentration, we have:

$$[\text{A}^+] = [\text{B}^-] \quad (2.23)$$

As another example, a molecule of the form A_2B has the following equilibrium in solution:



Charge balance requires:

$$\text{number of } A^+ \text{ particles} = 2 \times \text{number of } B^{2-} \text{ particles} \quad (2.24)$$

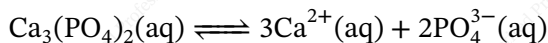
Therefore:

$$[A^+] = 2[B^{2-}] \quad (2.25)$$

The case above can be generalized to a solution containing multiple substances. Suppose the solution contains three substances, A_2B , AC , and A_3D , and assume that each salt dissociates completely without hydrolysis or other side reactions; each dissociates according to its own equilibrium equation, and the overall electrical neutrality of the solution requires the following equation to hold:

$$[A^+] = 2[B^{2-}] + [C^-] + 3[D^{3-}] \quad (2.26)$$

Let us look at a concrete example. To illustrate how a charge balance is set up, we assume that the compound $Ca_3(PO_4)_2$ dissociates completely without hydrolysis (an idealized treatment), giving the following dissociation equilibrium in aqueous solution:



According to the amount-of-substance balance relationship, we can directly obtain:

$$1 \times n(3Ca^{2+}) = 1 \times n(2PO_4^{3-}) \quad (2.27)$$

Since:

$$[(3Ca^{2+})] = \frac{1}{3}[Ca^{2+}] \quad (2.28)$$

$$[(2\text{PO}_4^{3-})] = \frac{1}{2} [\text{PO}_4^{3-}] \quad (2.29)$$

Therefore, for the aqueous system of $\text{Ca}_3(\text{PO}_4)_2$, the following charge balance relationship holds:

$$2 \times [\text{Ca}^{2+}] = 3 \times [\text{PO}_4^{3-}] \quad (2.30)$$

Under the idealized condition of complete dissociation, the physical meaning of this equation is: the total positive charge introduced by the solute equals the total negative charge introduced by the solute. It should be noted that for a real solution the charge balance equation must also include H^+ , OH^- , and the species produced by hydrolysis; the specific forms will be given together with worked examples in the “acid-base titration” chapter.

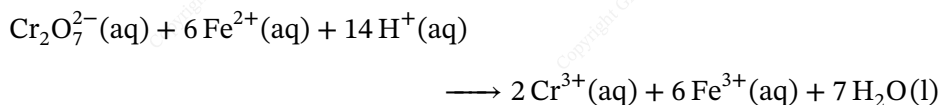
Stop and Think

If $\text{Al}_2(\text{SO}_4)_3$ is dissolved in pure water, what should the charge balance equation be? Hint: it dissociates into Al^{3+} and SO_4^{2-} , carrying charges of 3+ and 2−, respectively.

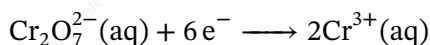
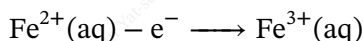
In dealing with relevant problems in acid-base titration systems, the charge balance method shows its unique advantages. We will demonstrate the practical application of this method in the “acid-base titration” chapter (Chapter ??).

2.4.4 Conservation of Electrons

The salient feature of a redox reaction is the transfer of electrons. For example, in acid solution, the dichromate ion oxidizes the ferrous ion as follows:



In this reaction, Fe^{2+} loses electrons and is oxidized to Fe^{3+} , while the chromium (oxidation state +6) in $\text{Cr}_2\text{O}_7^{2-}$ gains electrons and is reduced to Cr^{3+} . As for the specific mechanism of the electron transfer, it is not discussed here. What is certain is that the total number of electrons never changes: electrons can only be transferred from one substance to another, neither created out of nothing nor vanishing into nothing. The electron gain and loss of the relevant components in the above reaction is as follows (the half-reactions below show only the number of electrons gained or lost; for brevity the remaining components such as H^+ and H_2O are not filled in, and the complete balancing of half-reactions will be introduced in the redox titration chapter):



If the amount of substance of Fe^{2+} is $n(\text{Fe}^{2+})$ mol, then after complete reaction $n(\text{Fe}^{2+})$ mol of electrons are transferred out. Similarly, if the amount of substance of the dichromate ion is $n(\text{Cr}_2\text{O}_7^{2-})$ mol, then after complete reaction $6n(\text{Cr}_2\text{O}_7^{2-})$ mol of electrons are transferred in. Since in this reaction the gain and loss of electrons occurs only between these two species and the total count must be the same, we have:

$$1 \times n(\text{Fe}^{2+}) = 6 \times n(\text{Cr}_2\text{O}_7^{2-}) \quad (2.31)$$

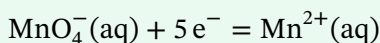
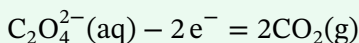
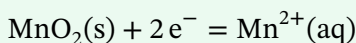
Common Pitfall

Conservation of electrons uses the number of electrons **gained or lost by each species**, not the coefficient in front of the reaction equation. A common mistake is to write $n(\text{Fe}^{2+}) = 1 \times n(\text{Cr}_2\text{O}_7^{2-})$ upon seeing that the coefficient in front of $\text{Cr}_2\text{O}_7^{2-}$ is 1. The correct approach is: first write the half-reactions to determine the number of electrons transferred by each species ($\text{Cr}_2\text{O}_7^{2-}$ gains 6, Fe^{2+} loses 1), then set up the equation by requiring the total numbers of electrons to be equal.

Equation 2.31 is the conservation of electrons. Using the law of conservation of electrons, the relationships between the amounts of substance of the relevant components in a redox reaction system can be established quickly. See the following example.

Example 2.7 A 0.2500 g sample containing MnO_2 is weighed, and dilute sulfuric acid and 0.3800 g of sodium oxalate are added and heated until the reaction is complete. The excess sodium oxalate is treated with a 0.03000 mol/L potassium permanganate solution, consuming 30.00 mL. Find the mass percentage of MnO_2 in the sample.

Solution: The gain and loss of electrons of the relevant components is as follows:



In the entire reaction, the species that loses electrons is $\text{C}_2\text{O}_4^{2-}$, while the species that gain electrons are MnO_2 and MnO_4^- , so:

$$2 \times n(\text{Na}_2\text{C}_2\text{O}_4) = 2 \times n(\text{MnO}_2) + 5 \times n(\text{KMnO}_4)$$

Therefore:

$$\begin{aligned}n(\text{MnO}_2) &= n(\text{Na}_2\text{C}_2\text{O}_4) - \frac{5}{2} \times n(\text{KMnO}_4) \\&= \frac{0.3800 \text{ g}}{134.00 \text{ g/mol}} - \frac{5}{2} \times 30.00 \times 10^{-3} \text{ L} \times 0.03000 \text{ mol/L} \\&= 5.858 \times 10^{-4} \text{ mol}\end{aligned}$$

The mass fraction of manganese dioxide is:

$$\begin{aligned}w(\text{MnO}_2) &= \frac{5.858 \times 10^{-4} \text{ mol} \times 86.94 \text{ g/mol}}{0.2500 \text{ g}} \times 100\% \\&= 20.37\%\end{aligned}$$

The stoichiometric balance relationships are the theoretical core of chemical quantitative analysis. Whether it is amount-of-substance balance, elemental balance, charge balance, or conservation of electrons, they jointly build the bridge of quantitative relationships in a reacting system, laying the foundation for subsequent titration analysis and the calculation of complex systems. Among them, the charge balance method will be applied systematically in the acid-base titration chapter (Chapter ??), and the conservation of electrons will play the same role in the redox titration chapter (Chapter ??).

Self-Check

1. $2[\text{Ca}^{2+}] = 3[\text{PO}_4^{3-}]$ belongs to _____ balance.
2. $n(\text{Fe}^{2+}) = 6n(\text{Cr}_2\text{O}_7^{2-})$ belongs to the conservation of _____.
3. $1n(\text{CH}_4) = n(\text{CO}_2)$ belongs to _____ balance.
4. When determining the calcium content of limestone, the sample is first dissolved with an excess of HCl, and the remaining HCl is back-titrated with a NaOH standard solution. Which balance is the most direct for establish-

ing the stoichiometric relationship between CaCO_3 and HCl ? What about between HCl and NaOH ? If the solution also contains a Mg^{2+} impurity that also consumes HCl , how should you deal with it?

Key Concept

Principles for selecting among the four stoichiometric balances: for an already balanced reaction use **amount-of-substance balance**; when we do not wish to balance the equation use **elemental balance**; for a solution system with multiple species use **charge balance**; and for a redox reaction use **conservation of electrons**. These four balance relationships are in essence conservation laws (conservation of atoms, conservation of charge, and conservation of electrons gained and lost) manifested in different scenarios, and choosing the correct type of balance is very important for efficient calculation.

2.5 Basic Concepts of Titrimetric Analysis

Titrimetric analysis (also called volumetric analysis) is one of the most widely applied methods in chemical quantitative analysis. Its essence is to determine the content of the analyte by measuring the volume of the standard solution consumed in the complete reaction with the analyte. With its advantages of simple equipment, convenient operation, and accurate results, this method plays an irreplaceable role in scientific research and production practice. This section systematically introduces the core framework shared by titrimetric analysis, such as the preparation and standardization of standard solutions, the four titration modes, and titration curves and indicators, laying a unified foundation for the in-depth study of the various specific titration methods in the chapters that follow.

2.5.1 The Titration Process and Terminology

The core operation of titrimetric analysis is titration. Titration refers to the process of adding a reagent of known accurate concentration, through a device called a burette, to the solution containing the analyte. This reagent of known accurate concentration is called the titrant or standard solution, as detailed in Section 2.5.2. The titrant is added drop by drop, which is also the origin of the term “titration.”

As the titrant is added, the concentration of the analyte in the system changes, and this changing process is usually displayed graphically, called the titration curve (see the titration-curve figures in Chapter ??). The horizontal axis of the titration curve is usually the amount of titrant consumed, while the vertical axis is commonly the negative logarithm of the equilibrium concentration of the analyte, such as pH. Near the equivalence point, the addition of only a very small amount of titrant causes the equilibrium concentration of the analyte to decrease sharply within a very small range of added volume, and its negative logarithm undergoes a corresponding sharp jump; this steep segment of the titration curve is called the titration break. The turning center of the titration break corresponds to the equivalence point of the titration, at which the ratio of the amount of titrant added to the amount of the analyte is exactly equal to the stoichiometric ratio of the reaction.

In theory, if the position of the equivalence point is known, the amount of the analyte can be calculated from the amount of titrant added and the stoichiometric relationship. However, because the titration process itself cannot reveal this position, a substance called an indicator must be introduced from outside to detect it by means of a change in its color. When the color of the indicator changes abruptly, the titration process is terminated, and the point at which the titration is

stopped is called the end point . For a real system, the position at which the color of the indicator changes often does not coincide exactly with the position of the equivalence point, and the deviation between the end point and the equivalence point gives rise to the titration error .

2.5.2 Standard Solutions

A standard solution is a solution whose component and concentration are both known. Titrimetric analysis requires the use of standard solutions, which in essence is equivalent to the need for weights in gravimetric analysis. In this sense, titrimetric analysis is essentially an extension of gravimetric analysis into the solution system.

In titrimetric analysis, a substance that can be used directly to prepare a standard solution is called a primary reagent, and it should satisfy the following requirements:

1. The composition of the primary reagent should agree exactly with its chemical formula. If the substance contains water of crystallization, the content of the water of crystallization should agree with the chemical formula;
2. The content of the main component of the primary reagent should be above 99.9%;
3. The primary reagent should have good stability, should not readily react with substances in the air, and should not readily adsorb substances from the air;
4. The molar mass of the primary reagent should be large.

Stop and Think

Suppose the weighing error of the analytical balance is ± 0.2 mg. To prepare 1.000 L of a standard solution of concentration 0.01000 mol/L, a total of 0.01000 mol of the primary reagent must be weighed out. (1) If the molar mass of the primary reagent is 10 g/mol, how many grams must be weighed out, and what is the relative error introduced by weighing? (2) What if a primary reagent of molar mass 100 g/mol is used instead? (3) From this calculation, can you summarize why a primary reagent requires a large molar mass?

Commonly used primary reagents can be divided, according to their chemical nature, into pure metals and pure compounds, as detailed in Appendix ??.

The methods commonly used to prepare a standard solution are the direct method and the standardization method. In the direct method, a certain amount of the primary reagent is accurately weighed with a balance, dissolved in an appropriate amount of water, then transferred completely into a volumetric flask and made up to the mark. The concentration of the standard solution is calculated from the mass weighed out and the volume of the volumetric flask. The standardization method is used in cases where the direct method cannot proceed, usually because the reagent cannot satisfy the requirements of a primary reagent. For example, the exact content of HCl in commercial hydrochloric acid is difficult to determine, so it cannot be directly made into a standard solution. To prepare a standard solution of HCl, one may first dilute the concentrated hydrochloric acid solution to about the desired concentration, then standardize it with borax, so as to obtain the accurate concentration of this hydrochloric acid solution.

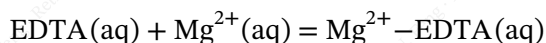
2.5.3 Requirements for Chemical Reactions and Common Titration Modes

In theory, any chemical reaction with a definite stoichiometric relationship can be used for titration analysis. However, only certain chemical reactions are ultimately used to construct a feasible titration analysis method. These reactions must satisfy the following four conditions:

1. There must be a definite stoichiometric relationship between the titrant and the analyte; otherwise the amount of the analyte cannot be calculated from the amount of titrant consumed;
2. The reaction must proceed sufficiently completely; near the equivalence point the degree of completion should generally exceed 99.9%, for otherwise the end point (the color change of the indicator) will deviate appreciably from the equivalence point and the titration error can hardly be kept within 0.1%;
3. The reaction rate must be fast enough, or it can be accelerated by optimizing the experimental conditions (such as heating, adding a catalyst), to meet the real-time requirement of the titration operation. If the reaction itself is very slow, it is easy to cause an excess of titrant, thereby producing a positive error;
4. There must be an appropriate and reliable means to indicate the end point of the titration.

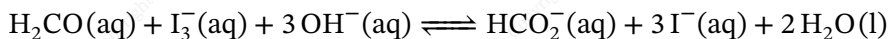
In order to maximize the scope of application of titration analysis, four basic titration modes have been developed for analytes of different natures and reaction characteristics, namely the direct titration method, the back titration method, the

replacement titration method, and the indirect titration method. Any chemical reaction that can satisfy the four conditions above can be used for quantitative analysis by the direct titration method. For example, EDTA reacts with magnesium ion as follows:



The stoichiometry of this reaction is a 1 : 1 relationship, the reaction rate is also very fast, and eriochrome black T can be used as an indicator to indicate the end point. Hence EDTA can be used for the direct titration of the magnesium ion.

If an analyte cannot be determined by the direct titration method, then, according to the specific circumstances, other titration modes may be used. For example, the formaldehyde content of a sample can be measured by titration analysis, and the main reaction involved is:



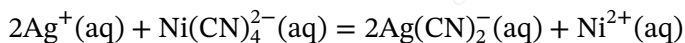
However, this is a slow reaction, so I_3^- cannot be used directly for titration analysis; in this case the back titration method can be adopted. First, a known amount of excess I_3^- is added, and the solution is allowed to stand under alkaline conditions for some time so that the reaction proceeds to completion; the solution is then acidified, and the excess iodine is titrated with a sodium thiosulfate standard solution.

The core principle of the back titration method is first to leave one reagent in excess, then to titrate the excess of that reagent. Another example of the back titration method is the titration of aluminum ion with EDTA. Because this reaction itself is very slow at room temperature, the aluminum ion blocks most metal indicators, and the aluminum ion may also undergo complex hydrolysis reactions.

Therefore, in titrating the aluminum ion, an excess of a known amount of EDTA standard solution is first added, the acidity is adjusted to pH 3.5, and the solution is boiled so that the aluminum ion is completely converted into a stable EDTA–Al complex; after cooling, the solution is adjusted to pH 5 ~ 6, and the excess EDTA is then back-titrated with a zinc ion standard solution, from which the content of the aluminum ion is calculated.

Sometimes the compound formed by the reaction of the analyte with a commonly used titrant is unstable (or has no definite stoichiometric relationship); in this case neither the direct titration method nor the back titration method can be used to determine the substance. However, with the aid of a replacement reaction, an equal amount of another substance can be displaced from another compound, and this substance can be determined by the direct titration method; in this way the determination of the original analyte is achieved. This is the replacement titration method.

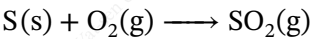
For example, the complex formed by the silver ion with EDTA is unstable and cannot be determined directly. In this case, $\text{Ni}(\text{CN})_4^{2-}$ can be added to the silver solution, and the following replacement reaction occurs:



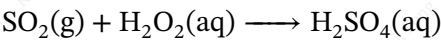
The Ni^{2+} thus released can be titrated with an EDTA standard solution, from which the content of the silver ion can be calculated.

There is another case where certain substances appear on the surface to be unable to be quantitatively analyzed by titration analysis, but they can be converted step by step through chemical reactions into a substance that can be determined by titration. For example, to determine the sulfur content of a sample, the direct titration method cannot be used, so the indirect titration method can be adopted.

First, the sample is burned so that the sulfur is converted into sulfur dioxide:



The sulfur dioxide produced is treated with hydrogen peroxide:



The sulfuric acid produced can be determined with a sodium hydroxide standard solution, from which the sulfur content of the sample can be calculated.

The applicable conditions and typical examples of the four titration modes can be summarized as follows:

Table 2.2: Comparison and selection of the four titration modes

Titration mode	Applicable conditions	Typical example
Direct titration	Complete reaction, definite stoichiometry, fast, end point can be indicated	EDTA directly titrating Mg^{2+}
Back titration	Slow reaction, or the analyte blocks the indicator	Add excess I_3^- to oxidize formaldehyde, then back-titrate with $\text{S}_2\text{O}_3^{2-}$
Replacement titration	Unstable product or no definite stoichiometric relationship	Ag^+ displacing $\text{Ni}(\text{CN})_4^{2-}$ to release Ni^{2+} , then titrating with EDTA
Indirect titration	The analyte does not react directly with the titrant	Sulfur burns $\rightarrow \text{SO}_2 \rightarrow \text{H}_2\text{SO}_4$, then titrated with NaOH

In practical application, the four titration modes have no absolute superiority or inferiority; the choice depends on the chemical nature of the analyte and the reaction conditions. The subsequent chapters (Chapter ?? through Chapter ??) will

separately demonstrate the application of these four modes with specific titration systems.

Common Pitfall

In the back titration method, the amount of standard solution consumed is not equal to the amount of the analyte; one must use “the total amount added” minus “the amount consumed in the back-titration” to obtain the amount that actually reacts with the analyte. Written as a formula: $n_{\text{reacted}} = c_{\text{total}}V_{\text{total}} - c_{\text{back}}V_{\text{back}}$. Forgetting this subtraction is the most common error in back-titration calculations.

Stop and Think

(1) 50.00 mL of 0.02000 mol/L EDTA is added to a solution containing Al^{3+} , and the back-titration consumes 15.00 mL of a 0.01000 mol/L Zn^{2+} . How many millimoles of EDTA reacted with Al^{3+} ?

(2) Both the back titration method and the indirect titration method involve multiple steps of reaction, but they differ in essence. In the back titration method, the first step deliberately adds an **excess** of reagent, which is then back-titrated; in the indirect titration method the analyte itself **does not react directly with the titrant** and must first be converted into a determinable substance. Describe the “first step operation” of the two methods in one sentence each. Can you see the fundamental difference between them?

2.5.4 Characteristics and Classification of Titrimetric Analysis

As a classical quantitative means, titrimetric analysis has the outstanding advantages of simple equipment, low cost, convenient operation, and high accuracy (the relative error can reach about 0.1%; for example, the reading error of a 50 mL burette is about ± 0.02 mL, giving a relative error of only 0.04%), so that it occupies a core position in the analysis of major constituents. Its main limitation is that

it is usually applicable to constituents of relatively high content (generally $> 1\%$) and it is lower in sensitivity than modern instrumental analysis methods.

According to the type of chemical reaction on which the titration reaction is based, titrimetric analysis can be divided into four major classes:

- Acid-base titration — based on proton-transfer reactions.
- Complexation titration — based on complexation reactions, of which the complexation titration with EDTA as the titrant is the most widely applied.
- Redox titration — based on electron-transfer reactions; according to the titrant used, it can be divided into the permanganate method, the dichromate method, the iodometric method, and so on.
- Precipitation titration — based on precipitation-forming reactions, such as the argentometric method.

These four classes constitute the main body of titrimetric analysis, and each has its own distinctive theoretical system, method of indicating the end point, and scope of application. The subsequent chapters will investigate these methods in depth. Understanding the general principles and concepts introduced in this section is the key to entering the study of each specialized field.

Key Concept

The core idea of chemical quantitative analysis can be summarized as follows: **taking a precisely measurable physical quantity (mass or volume) as the bridge, through the rigorous stoichiometric relationship of a chemical reaction, deducing the content of the analyte.** Gravimetric analysis directly weighs the reaction product or the loss in mass of the sample, while titrimetric analysis achieves the same purpose indirectly through volume measurement.

Having understood this core idea, you can answer a more fundamental question: **why should one trust an analytical result?** The answer lies at the start of the traceability chain, namely gravimetry reads directly a mass, and mass can be traced to the SI definition of the kilogram. The numbers given by instrumental analysis (absorbance, peak area), after passing through the calibration curve, the standard solution, the standardization of the primary reagent, and many other links, must all eventually connect to this starting point. If you obtain a datum in the laboratory but cannot state clearly by which pathway it connects to this starting point, then the credibility of this datum is in question.

2.6 General Steps of Chemical Quantitative Analysis

Chemical quantitative analysis is a systematic process consisting of several interrelated steps. The complete analytical workflow can be divided into four closely connected steps: sampling, sample preparation, measurement, and the expression and evaluation of the results. The steps are interlocking, and the quality of the operation in any one step will directly determine the accuracy and reliability of the final analytical result.

2.6.1 Sampling

Analytical sampling is the first step in carrying out a specific analytical procedure, and its accuracy and representativeness directly affect the reliability of the final analytical result. In practical quantitative analysis, the kinds of samples encountered are numerous and the amount of sample varies greatly; it may be extremely limited, or it may be quite ample. When the amount of sample is small, selective sampling usually cannot be carried out, and the whole sample must be used for analysis. However, if the total amount of sample is too large, it is obviously inappropriate to bring the entire sample into the analytical workflow, for otherwise the cost of analysis would be too high and the efficiency lowered. Therefore, according to the property and state of the sample, an appropriate and representative portion must be scientifically collected from the provided sample for laboratory analysis.

In most cases, the amount of sample actually collected is far smaller than the total amount of the original sample provided. Yet the analytical task requires that the determination of this small portion of the sample accurately reflect the true information of the original whole, so the representativeness of the collected sample is crucial. For homogeneous samples such as solutions or gases, provided the whole is uniformly mixed, the portion taken from it usually has good representativeness, and such cases are not discussed further here. For solid samples, one of the common methods of obtaining a representative sample is random sampling, whose basic mode of operation is shown in Figure 2.4.

It should be noted here that random sampling is not arbitrary sampling, but a systematic mode of sampling guided by the theory of mathematical statistics in accordance with the principle of randomness. Taking Figure 2.4 as an example, the sampling positions shown are scientifically deployed points on a planar region

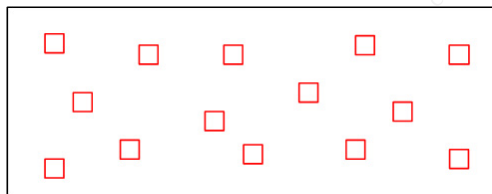


Figure 2.4: Schematic diagram of random sampling

on the basis of two-dimensional random numbers. For a bulk sample composed of many independent units (such as a whole trainload of iron ore), sampling at only one position is clearly insufficient to represent the whole; in such a case a multi-stage sampling strategy must be adopted, in which the samples are taken independently and randomly from different units, so as to improve the representativeness of the sample.

Besides the choice of the sampling method, the determination of the amount of sample is likewise crucial. The amount sampled must be strictly calculated and scientifically designed, and cannot be set arbitrarily. Statistical research clearly shows that if the uncertainty introduced by the sampling process is too large, it will significantly affect the accuracy of the final determination and may even lead to a fundamentally wrong conclusion. Because the mathematical basis involved in sampling theory goes beyond the scope of this book, it is not developed further here. It must be emphasized in particular that, in quantitative analysis, **the sampling error is often the largest contributor to the total error**. Suppose the true standard deviation of the iron content of a batch of iron ore is 0.20%, while the standard deviation of the laboratory measurement step is only 0.02%; then the total uncertainty is almost entirely determined by the sampling step. Even if the precision of the measurement is improved tenfold, as long as the sampling is not improved, the reliability of the final result will not increase. This is why sampling must not be treated casually: it is the weakest link of the entire analytical chain,

and the link most easily overlooked.

Stop and Think

From a batch of 500 kg of ore, you take only 1 g of sample to analyze the iron content. If the iron distribution in the ore is extremely non-uniform (some particles are pure Fe_2O_3 , some pure SiO_2), can this 1 g of sample represent the 500 kg? If not, what do you think should be done at the very least?

2.6.2 Sample Preparation

The sample obtained from sampling is usually called the gross sample. After it is transported to the laboratory, it must be further prepared into a laboratory sample that meets the analytical requirements. For a solid sample, the preparation of the laboratory sample usually includes steps such as crushing and reduction (sample splitting). The reason is that the solid sample collected is usually not uniform, and the amount of sample is relatively large, so it generally cannot be used directly in the measurement step.

The first step of sample preparation is to crush and grind the gross sample many times and to control the particle size by sieving, so as to improve the uniformity of the sample. The ground and uniform sample must then be reduced in order to obtain the minimum amount of sample suitable for analysis. The commonly used reduction method is the quartering method, whose procedure is: mix the sample, pile it into a cone, flatten it into a round cake, then divide it into four equal parts through the center, keep the two opposite parts and discard the other two; repeat this operation until the required amount of sample is obtained. Figure 2.5 shows the general process of the preparation of a solid sample.

The sample preparation process described above may require several cycles to reach the requirement, and the minimum mass of the final sample can be esti-

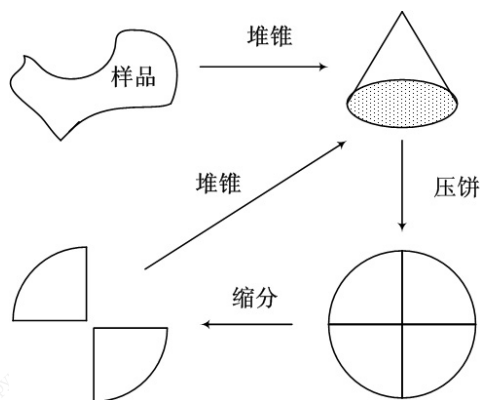


Figure 2.5: The general process of sample preparation

related with the following formula:

$$m_Q \geq kd^2 \quad (2.32)$$

where m_Q is the minimum mass of the required sample (unit kg), d is the maximum particle size of the sample (unit mm), and k is the reduction constant, which usually takes values in the range $0.05 \sim 1 \text{ kg/mm}^2$.

Stop and Think

If the particle size is reduced to 1/10 of the original, to what fraction is the allowed minimum amount of sample reduced? Why is crushing given priority over reduction?

Example 2.8 There is a sample of 30 kg whose final particle size is required to be no larger than 2.00 mm. How many times should it be reduced? It is known that $k = 0.3 \text{ kg/mm}^2$.

Solution: The minimum mass is:

$$m_Q \geq kd^2 = 0.3 \text{ kg/mm}^2 \times (2.00 \text{ mm})^2 = 1.2 \text{ kg}$$

The amount of sample remaining after four reductions is $30 \text{ kg} \times 0.5^4 = 1.875 \text{ kg}$, which is still greater than 1.2 kg; the amount remaining after five reductions would be $30 \text{ kg} \times 0.5^5 = 0.9375 \text{ kg}$, below 1.2 kg; hence the fifth reduction is not allowed, and the sample should be reduced four times.

The sample obtained from the preparation process described above usually contains the analyte together with other impurities, and its existing forms are complex, so it often cannot be used directly for quantitative analysis. Therefore, the sample must be pretreated before analysis, so that the analyte is transferred quantitatively into solution and potential interferences are eliminated as far as possible.

A complete sample pretreatment usually comprises several steps: first, the analyte is dissolved, fused, or extracted out of the sample matrix by digestion or extraction; then a separation and cleanup step removes the influence of the matrix and interfering substances; when necessary, the analyte is enriched by concentration so as to raise its concentration, or is converted by derivatization into a species that is easier to determine. Every step of the pretreatment is decisive for the accuracy of the analytical result. This section focuses on the first of these steps, digestion and extraction; the common methods and techniques are summarized in Figure 2.6.

As shown in Figure 2.6, the digestion method for an inorganic sample is chosen according to the physical state and the volume of the sample. For inorganic solid samples, the common methods fall into three classes: electric-heating digestion, dry fusion digestion, and microwave digestion. Electric-heating digestion is carried out at atmospheric pressure on an electric hot plate, with water, acid, base, or mixed acids as the solvent, and the sample is dissolved through acid-base

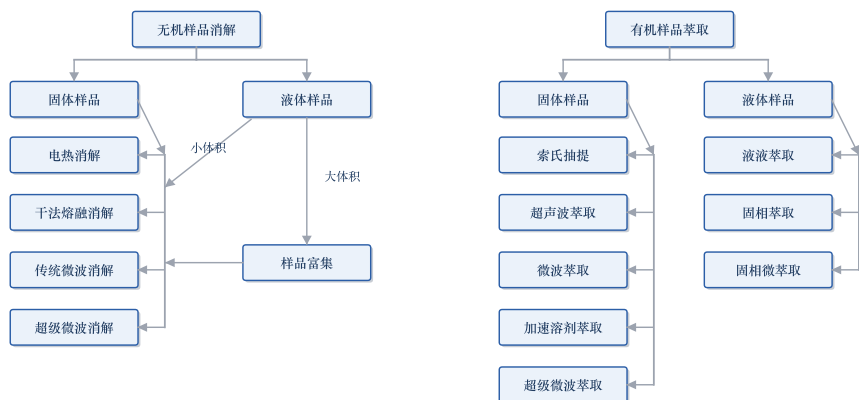


Figure 2.6: Methods and techniques for sample digestion and extraction

or redox reactions. For example, hydrochloric acid can be used to dissolve pure metals and ores whose major constituents are alkaline-earth metals. For refractory matrices that such solvents can hardly decompose, such as silicates, dry fusion digestion can be used: a solid flux is fused together with the sample at high temperature, converting it into products soluble in water or acid. For example, in determining the silicon content of a soil, KOH is fused with the soil sample, the fused mass is dissolved in acid, and the solution is then subjected to titration analysis. Microwave digestion heats the acid system with microwaves in a closed vessel and intensifies the dissolution by raising both the temperature and the pressure; it combines the advantages of high speed, low acid consumption, and small loss of volatile components.

For inorganic liquid samples, the treatment depends on the sample volume. A small-volume sample can be digested directly, whereas a large-volume sample must first be enriched by concentration, coprecipitation, or solid-phase extraction before digestion, so that the concentration of the analyte reaches the detection-limit requirement of the analytical method.

The strategy for organic samples differs from that for inorganic samples: if an

organic constituent is to be determined, the analyte must be extracted out of the matrix and retained; if an inorganic constituent is to be determined, the organic matrix must be decomposed and removed, usually by dry ashing or wet digestion.

For organic solid samples, the classical method is Soxhlet extraction: the solvent is refluxed and condensed repeatedly in the extractor, continuously extracting over a long period of time; the results are reliable, but the method takes several hours and consumes large amounts of solvent. Modern techniques markedly shorten the extraction time by introducing intensified means: ultrasonic extraction promotes mass transfer by the cavitation effect; microwave extraction exploits the in-situ heating of polar molecules (water, polar solvents) in a microwave field, so that the analyte is released quickly; accelerated solvent extraction keeps the solvent in the liquid state at high temperature and pressure and achieves rapid extraction through the increased solubility and diffusion rate. These methods can shorten the extraction time from several hours to tens of minutes or less, and reduce the solvent consumption by more than an order of magnitude.

For organic liquid samples, liquid-liquid extraction, solid-phase extraction, and solid-phase microextraction are commonly used. Liquid-liquid extraction is based on the partition of the analyte between an aqueous phase and an organic phase, and is the classical method with the longest history and the widest application; solid-phase extraction replaces the extraction solvent with a solid sorbent and integrates extraction, cleanup, and enrichment in one step, and is easy to automate; solid-phase microextraction uses a quartz fiber coated with a stationary phase to adsorb the analyte directly from the sample, after which the fiber can be introduced directly for desorption and analysis, using almost no organic solvent and meeting the requirements of green analytical chemistry.

It is worth mentioning that microwave pretreatment based on a pre-

pressurized single-reaction-chamber system (Single Reaction Chamber, SRC), commonly known as “super microwave,” has developed rapidly in recent years. Its working temperature and pressure can reach as high as 300°C and 20 MPa, respectively, and it features strong batch-processing capability, high efficiency, and low operating cost. The technology can be used both for the digestion of inorganic constituents and for the extraction of organic constituents, and represents the development direction of sample pretreatment. Figure 2.7 shows a commercial super-microwave pretreatment instrument.



Figure 2.7: A Boao super-microwave digestion/extraction system. Image from the website of Guangdong Boao Scientific Instruments Co., Ltd.

2.6.3 Measurement of the Sample

In actual analysis, the measurement of the sample should take into account both the accuracy and the speed. In general, the higher the content of the analyte, the higher the accuracy required of the analytical result. For example:

- When the content falls in the high-concentration range 50% ~ 100%, the rel-

ative error of the measurement is usually required to be controlled within 0.1% ~ 0.3%, in which case classical methods such as gravimetric analysis and titrimetric analysis can be used to meet the higher accuracy requirement.

- When the content falls in the lower-concentration range 0.01% ~ 1%, the accuracy requirement may be relaxed accordingly, and the permitted relative error is generally 2% ~ 10%; in this case instrumental analysis methods, which have high sensitivity and are suitable for trace analysis, are more appropriate.

The relationship above is summarized as follows:

Method type	Relative error	Concentration range	Speed of analysis
Gravimetry/Titrimetry (classical)	0.1% ~ 0.3%	1% ~ 100%	relatively slow
Instrumental analysis	2% ~ 10%	0.01% ~ 1%	fast

This table is not absolute but merely a typical reference range. The accuracy of some instrumental analysis methods can also reach 0.5%, and some titration methods under optimized conditions can also be used for the determination of trace constituents. In choosing an analytical method, the accuracy requirement, the concentration range, the speed of analysis, and the cost should be considered together.

Besides accuracy, the speed of analysis is also an important factor that must be considered in actual work. Taking a modern large chemical plant as an example, its production process runs mostly continuously, and a fluctuation in the quality of an intermediate product at any stage may affect the stable operation of the whole process. Therefore, fast and reliable analytical methods are needed to

complete the determination of the components of an intermediate product or finished product in a short time, so as to adjust the process parameters in time and ensure the continuity of production and the stability of product quality.

Stop and Think

A pharmaceutical plant needs to test the purity of a newly synthesized drug, requiring a relative error $\leq 0.3\%$ and a content of about 98%. Another laboratory needs to determine the lead content of river water, about 0.005 mg/L. For these two tasks, should a chemical analysis method or an instrumental analysis method be chosen? Why?

2.6.4 Expression and Evaluation of Results

In actual analysis, the chemical form of the analyte in the sample may differ from the form at the final measurement, so the manner of expressing the analytical result must be reasonably selected according to the actual situation. Taking the analysis of iron ore as an example, the iron element in it may exist in various oxide forms such as Fe_2O_3 and FeO , while in the measurement process all the forms of iron are often converted into Fe^{3+} before determination. At this point, whether the final result should be reported in the form of Fe_2O_3 or FeO becomes a typical problem of the manner of expression. Such problems are widespread in chemical quantitative analysis and are usually determined in combination with the specific purpose of the analysis, industry practice, or the relevant standard.

Besides the choice of the manner of expression described above, the specific expression of the analytical result also depends on the physical state of the sample. For solid samples, the analytical result is usually expressed as the mass fraction, as follows:

$$w_{\text{B}} = \frac{m_{\text{B}}}{m_{\text{S}}} \quad (2.33)$$

where w_B denotes the mass fraction of the analyte B; m_B is the mass of the analyte; and m_S is the mass of the sample.

When the units of m_B and m_S are the same, w_B is usually expressed as a percentage. When the content of the target constituent is very low, $\mu\text{g/g}$, ng/g , and pg/g can be used.

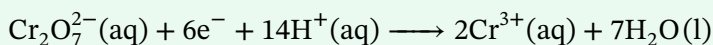
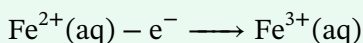
For liquid samples, the analytical result is usually expressed as the amount-of-substance concentration of the analyte, and alternatively the mass fraction or the volume fraction may be used as needed. For gas samples, the analytical result is usually expressed as the volume fraction.

Observed example, excerpt from a water-quality analysis report: In the 2023 water-quality inspection report of a city's tap-water company, the lead content is stated as $2.1\ \mu\text{g/L}$, and the cadmium content as $0.3\ \mu\text{g/L}$. Behind these two numbers lies a whole series of steps, from the deployment of sampling points (12 network-end points, sampled once each quarter), sample acidification and preservation ($\text{pH} < 2$), determination by graphite-furnace atomic absorption spectrometry, calibration-curve calibration, blank correction, and finally data rounding and unit conversion. **Try to trace:** in this report, which links involve the sampling, sample preparation, measurement, and expression of results that you have learned about in this chapter?

Example 2.9 A $0.5000\ \text{g}$ sample of iron ore is weighed, dissolved, and all the iron is reduced to ferrous ion. The complete reaction with a $0.02000\ \text{mol/L}$ potassium dichromate standard solution consumes $44.76\ \text{mL}$. Find the mass fraction of iron in the sample, expressing the re-

sult in the form of both Fe and Fe_2O_3 .

Solution: In this redox reaction, Fe^{2+} is oxidized to Fe^{3+} , and $\text{Cr}_2\text{O}_7^{2-}$ is reduced to Cr^{3+} . The respective half-reactions and the numbers of electrons transferred are:



(each Fe^{2+} loses 1 electron, and each $\text{Cr}_2\text{O}_7^{2-}$ gains 6 electrons)

According to the conservation of electrons, the total number of electrons lost by Fe^{2+} equals the total number of electrons gained by $\text{Cr}_2\text{O}_7^{2-}$:

$$1 \times n(\text{Fe}^{2+}) = 6 \times n(\text{K}_2\text{Cr}_2\text{O}_7)$$

That is:

$$\frac{m(\text{Fe}^{2+})}{M(\text{Fe})} = 6 \times c(\text{K}_2\text{Cr}_2\text{O}_7) \times V(\text{K}_2\text{Cr}_2\text{O}_7)$$

Therefore:

$$\begin{aligned} m(\text{Fe}^{2+}) &= 6 \times c(\text{K}_2\text{Cr}_2\text{O}_7) \times V(\text{K}_2\text{Cr}_2\text{O}_7) \times M(\text{Fe}) \\ &= 6 \times 0.02000 \text{ mol/L} \times 44.76 \times 10^{-3} \text{ L} \times 55.85 \text{ g/mol} \\ &= 0.3000 \text{ g} \end{aligned}$$

When the result is expressed in the form of Fe, the mass fraction is:

$$\begin{aligned} w(\text{Fe}) &= \frac{m(\text{Fe}^{2+})}{m_s} \times 100\% \\ &= \frac{0.3000 \text{ g}}{0.5000 \text{ g}} \times 100\% \\ &= 60.00\% \end{aligned}$$

When the result is expressed in the form of Fe_2O_3 , since for the same sample there is the following stoichiometric relationship:

$$n(\text{Fe}) = 2 \times n(\text{Fe}_2\text{O}_3)$$

the mass fraction is:

$$\begin{aligned} w(\text{Fe}_2\text{O}_3) &= \frac{n(\text{Fe}_2\text{O}_3) \times M(\text{Fe}_2\text{O}_3)}{m_s} \times 100\% \\ &= \frac{1}{2} \times \frac{m(\text{Fe}^{2+})}{M(\text{Fe})} \times \frac{M(\text{Fe}_2\text{O}_3)}{m_s} \times 100\% \\ &= \frac{1}{2} \times \frac{0.3000 \text{ g}}{55.85 \text{ g/mol}} \times \frac{159.7 \text{ g/mol}}{0.5000 \text{ g}} \times 100\% \\ &= 85.78\% \end{aligned}$$

This section has introduced the four main steps of chemical quantitative analysis. It should be emphasized that every step, from sampling, to sample preparation, to sample measurement, introduces errors, which are in the end transmitted and accumulated into the analytical result. How to evaluate the validity and reliability of the final result is likewise an indispensable step. These matters involve a considerable amount of theory, and we take them up in the chapter on data processing (Chapter ??).

Key Concept

The four main steps of analysis: sampling (representativeness is the core) → sample preparation (crushing, reduction, digestion) → measurement (choosing the classical or the instrumental method depends on the accuracy and the concentration) → expression of the result (reporting in the appropriate chemical form and unit). Every step introduces errors, and the total uncertainty of the final result is the accumulation of the errors of each step.

Chapter Summary

1. **Classification of chemical quantitative analysis methods.** They are divided into gravimetric analysis (with mass as the measurement signal) and titrimetric analysis (with volume as the measurement signal). Gravimetric analysis is the fundamental basis of the absolute quantitative methods, and all quantitative analysis methods must be traceable, in some manner, to gravimetric analysis.
2. **Basic physical quantities and concentration.** Mass, volume, relative atomic mass, relative molecular mass, amount of substance (mol), molar mass, and amount-of-substance concentration constitute the basic system of quantities and units of chemical quantitative analysis. The flexible construction of the “basic unit” is the core feature of the metrological system of this textbook: by adjusting the basic unit (for example $\frac{1}{2}\text{H}_2\text{SO}_4$ or 2KMnO_4), the stoichiometric coefficients of any reaction can be unified to 1 : 1, greatly simplifying the calculation.
3. **Analytical concentration vs. equilibrium concentration.** Analytical concentration c denotes the total amount of a substance in 1 L of solution (tracing back to the original species), while equilibrium concentration $[]$ denotes the concentrations of the actual species present in solution. The

relationship between the two is linked through the distribution fraction, and is the basis of the subsequent theories of acid-base equilibrium and complexation equilibrium.

4. **The four stoichiometric balance relationships.** Amount-of-substance balance (derived from the stoichiometric equation), elemental balance (conservation of atoms), charge balance (electrical neutrality of solution), and conservation of electrons (conservation of the number of electrons transferred) constitute the system of calculation methods of chemical quantitative analysis. They are the general tools of all the subsequent chapters.
5. **Sampling and sample preparation.** Sampling must follow the principle of randomness (not arbitrary sampling); solid samples are reduced by crushing, sieving, and the quartering method; the minimum mass of the final sample satisfies $m_Q \geq kd^2$. Sample pretreatment methods depend on the type of sample: inorganic samples are commonly treated by electric-heating digestion, microwave digestion, or fusion; organic samples may be extracted by Soxhlet extraction, ultrasonic extraction, microwave extraction, or accelerated solvent extraction, or have their organic matrix removed by dry ashing or wet digestion.
6. **Core concepts of titration analysis.** Titrant (standard solution) → equivalence point (theoretical equivalence point) → end point (the point of indicator color change) → titration error (deviation between the end point and the equivalence point). The accurate judgment of the end point is a necessary condition for an accurate titration.
7. **Standard solutions and primary reagents.** Standard solutions are prepared by the direct method (directly prepared from a primary reagent) or the standardization method (standardizing an approximate concentration with a primary reagent). The primary reagent must satisfy the four requirements that its composition agrees with the chemical formula, its purity is

above 99.9%, it is stable, and its molar mass is large.

8. **The four titration modes.** Direct titration, back titration, replacement titration, and indirect titration are respectively applicable to different reaction conditions. In selecting a mode, direct titration should be preferred; when the reaction is slow, the product is unstable, or the titrant does not react directly with the analyte, the other modes are chosen in turn.
9. **The relationship between chemical quantitative analysis and instrumental analysis.** Gravimetric and titrimetric methods are the calibration cornerstone of modern instrumental analysis; almost all instrumental analysis methods must be traced through standard substances to these absolute methods. Chemical quantitative analysis is irreplaceable in standard calibration, high-accuracy determination of major constituents, emergency analysis when no standard reference material is available, and foundational training.

Self-Check

Self-study check: after reading this chapter, you should be able to:

Basic level (can you do it?)

- Distinguish gravimetric analysis from titrimetric analysis; state which physical quantity each uses as the measurement signal
- Starting from mass and volume, pass through relative atomic mass, amount of substance, and molar mass to finally calculate the amount-of-substance concentration; for each step, state which formula is being used
- In a given chemical reaction, construct the respective basic units so that all the stoichiometric coefficients become 1
- Independently complete the stoichiometric derivation for the ammonium sulfate-ammonium nitrate mixture
- Use the charge balance equation to constrain the concentrations of the var-

ious species in solution

- State the four requirements for a primary reagent and the reason behind each of them
- Given a reaction and the experimental conditions, determine the titration mode to be used (direct/back/replacement/indirect)
- From a blank burette reading to the final reported result, output a complete data chain
- Explain why instrumental analysis methods must ultimately be traced to gravimetry

In-depth level (can you explain it?)

- Explain to a classmate: why $1 \text{ mL} = 1 \text{ cm}^3$ holds exactly, while $1 \text{ L} = 1 \text{ dm}^3$ was only approximate historically?
- Among the four requirements for a primary reagent, use a specific numerical example to explain to a classmate the reason for the requirement “the molar mass must be large”
- The four stoichiometric balance relationships (amount-of-substance balance, elemental balance, charge balance, conservation of electrons): when faced with a new reacting system, how do you decide which one to use first?
- Someone tells you “titration is just measuring a volume, very simple.” Using what you have learned in this chapter, list at least three reasons to refute this claim

Exercises

2.1 What is a standard solution? What methods are available for preparing a standard solution?

2.2 What is a primary reagent? What conditions should it satisfy as a primary reagent?

★★ 2.3 What effect will the following situations have on the analytical result? A. positive error; B. negative error; C. no effect; D. cannot be determined.

1. When standardizing the concentration of a hydrochloric acid solution, the primary reagent Na_2CO_3 used contains a small amount of NaHCO_3 ;
2. After heating to dissolve the primary reagent, the solution is transferred to a volumetric flask without cooling and diluted to the mark, shaken, and immediately used for standardization;
3. When preparing a standard solution, the solution in the volumetric flask is not shaken uniform;
4. When transferring the sample solution with a pipette, the pipette is not rinsed with the solution to be transferred beforehand;

★★ 2.4 For the following scenarios, how would you collect a representative sample?

1. Water in a small stream.
2. Polymer film produced by a chemical plant.
3. Aspirin tablets in a medicine bottle.
4. Soil on the lawn between the north and south buildings of the chemistry college.
5. Old paint on the exterior wall of a building.
6. Animal tissue dissected to study an insecticide.

★★ 2.5 Explain the importance of reducing the particle size of a solid sample during its preparation.

★★ 2.6 Find the following concentrations:

1. 0.200 L of a sodium hydroxide solution contains 6.000 g of NaOH; find $c(\text{NaOH})$.

2. 1000. mL of a solution contains AgNO_3 3.398 g; find $c(\text{AgNO}_3)$.
3. Using 4.740 g of potassium permanganate to prepare 750. mL of solution, find $c(\frac{1}{5}\text{KMnO}_4)$.
4. Weigh 4.560 g of a $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ sample and prepare 500. mL of solution; find $c(\text{CuSO}_4)$.

★★ 2.7 In order to determine the sulfur content of a sample containing $\text{C}_{10}\text{H}_{20}\text{N}_2\text{S}_4$, 0.4613 g of the sample is weighed, and by an oxidation reaction all the sulfur in it is converted into sulfur dioxide; the sulfur dioxide is then passed into a solution containing hydrogen peroxide to form sulfuric acid; finally the sulfuric acid formed reacts completely with a 0.02500 mol/L sodium hydroxide solution, consuming 34.85 mL. Find the mass fraction of sulfur.

★★ 2.8 There is a potassium permanganate solution whose concentration is 0.04000 mol/L. To prepare a sodium oxalate solution such that it reacts completely with the above permanganate in equal volume under acid conditions, what should be the concentration of the sodium oxalate solution?

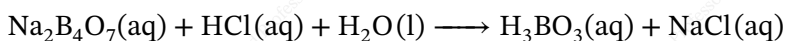
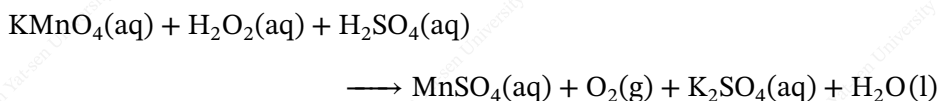
★★ 2.9 Weigh out exactly 1.2258 g of $\text{K}_2\text{Cr}_2\text{O}_7$, dissolve it in distilled water in a 150 mL beaker, then transfer it to a 250.0 mL volumetric flask and dilute to the mark with distilled water. Accurately transfer 25.00 mL of this solution to a conical flask, add HCl and an excess of KI solution, use starch solution as the indicator, and titrate to the end point with the $\text{Na}_2\text{S}_2\text{O}_3$ solution to be standardized, consuming 24.95 mL. Find the concentration of the $\text{Na}_2\text{S}_2\text{O}_3$ solution.

★★ 2.10 Weigh out about 1.700 g of the disodium salt of EDTA and prepare a solution. This solution reacts with 25.00 mL of a 0.01000 mol/L zinc ion solution, and the reaction is complete when 24.98 mL of EDTA is consumed. Find the concentration of the EDTA solution. The reaction of EDTA with Zn is a 1 : 1

relationship.

★★ 2.11 Weigh 0.4185 g of iron ore, dissolve it in acid, and use a reducing column to reduce all the ferric ions to Fe^{2+} . When this iron solution reacts completely with a 0.02500 mol/L potassium permanganate solution, the latter consumes 41.27 mL. Calculate the mass of iron in the iron ore.

★★ 2.12 Balance the following reaction equations, then write out the relationships between the amounts of substance of the components by both the traditional stoichiometric method and the method of unified basic units.



★★ 2.13 Dissolve 4.900 g of pure H_2SO_4 in water and make up to 1000.0 mL. Calculate:

1. $c(\text{H}_2\text{SO}_4)$ (mol/L);
2. $c\left(\frac{1}{2}\text{H}_2\text{SO}_4\right)$ (mol/L);
3. The analytical concentration of H^+ in this solution (assuming H_2SO_4 dissociates completely into H^+ and SO_4^{2-}).

Given $M(\text{H}_2\text{SO}_4) = 98.078 \text{ g/mol}$.

★★ 2.14 The following six analytes are given. For each, choose the most appropriate titration mode (one of direct titration, back titration, replacement titration, or indirect titration), and briefly state the reason.

1. The Ca^{2+} content of a water sample. Ca^{2+} reacts quantitatively with EDTA at $\text{pH} \approx 10$, with eriochrome black T as the indicator.

2. The Al^{3+} content of wastewater. Al^{3+} reacts extremely slowly with EDTA at room temperature, and Al^{3+} blocks the commonly used metal indicators.
3. The NaCl content of a foodstuff. Na^+ does not react directly with EDTA.
4. The SO_4^{2-} content of natural water. SO_4^{2-} does not coordinate directly with EDTA.
5. The Cr content of steel. First an oxidant is used to convert Cr into $\text{Cr}_2\text{O}_7^{2-}$, then an excess of Fe^{2+} standard solution is added, and finally $\text{K}_2\text{Cr}_2\text{O}_7$ is used for the back-titration.
6. The content of free HF in an electrolyte. HF is a weak acid, but the end point is not clear when titrated directly with NaOH.

★★★ 2.15 To determine the Zn content of an alloy, 0.5000 g of the sample is weighed, dissolved in acid, an ammonia buffer solution is added to adjust $\text{pH} = 10.00$, then 50.00 mL of a 0.02000 mol/L EDTA standard solution is added. It is heated so that all the metal ions that can coordinate react completely with EDTA. After cooling, with eriochrome black T as the indicator, the remaining EDTA is back-titrated with a 0.01000 mol/L Zn^{2+} standard solution, consuming 12.34 mL. Find the mass fraction of Zn in the alloy. Given $M(\text{Zn}) = 65.38 \text{ g/mol}$; assume that only Zn in the alloy reacts with EDTA.

★★★ 2.16 A batch of ore has a total mass of about 500 kg and a maximum particle size of about 25 mm. It is now to be prepared into a laboratory sample with particle size no larger than 2.00 mm.

1. If the reduction constant $k = 0.5 \text{ kg/mm}^2$, once the sample has been crushed to a particle size no larger than 2.00 mm (crushing does not change the sample mass), at most how many reductions can be carried

out in the whole process? (Each reduction halves the sample mass.)

2. If the sample is first crushed to a particle size below 5.00 mm and reduced a number of times, and is then crushed to a particle size below 2.00 mm and reduced further, at most how many reductions can be carried out in the whole process?

★★★ 2.17 A research group published a paper in *Analytical Chemistry* reporting the determination of sulfate ion content in river water by precipitation gravimetry. The experimental method was as follows: a 500.0 mL river-water sample was taken, and an excess of BaCl_2 solution was added to precipitate BaSO_4 . The precipitate was filtered, washed, dried, and then ignited at 800°C to constant mass. The final mass of the BaSO_4 precipitate obtained was 0.2356 g.

Questions: (1) What is the concentration of sulfate in the river water (in mg/L)? (2) Why is an excess of BaCl_2 added first and then ignited to constant mass, rather than titrating directly in a burette? (3) If the river water also contains CO_3^{2-} , what effect will this have on the result of this gravimetric determination? How should it be handled?

Given: $M(\text{BaSO}_4) = 233.39 \text{ g/mol}$, $M(\text{SO}_4^{2-}) = 96.06 \text{ g/mol}$

★★★ 2.18 A mining company sends a batch of alloy sample of unknown composition and asks you to determine the nickel content. It is known that the alloy is probably a nickel-cobalt alloy or a nickel-copper alloy, and the nickel content is roughly in the range 10% ~ 30%. Your laboratory has the following reagents and instruments: EDTA standard solution (0.02000 mol/L), KCN solution, AgNO_3 solution, HCl (6 mol/L), NaOH solution (2 mol/L), murexide indicator, a 50 mL burette, an analytical balance (sensitivity 0.0001 g), and a muffle furnace.

Please design a complete analytical scheme, including the following contents:

1. The weighing and dissolution steps of the sample;
2. The titration mode chosen (and the reason);
3. The choice of titrant and indicator (and the reason);
4. The masking or separation method for the interfering ions;
5. The whole analytical procedure given in the form of a flowchart.

★★★ 2.19 A student used the conservation of electrons to treat the reaction of MnO_4^- with Fe^{2+} in acid solution and wrote $n(\text{KMnO}_4) = 5n(\text{Fe}^{2+})$. Point out the error, write the correct relationship, and explain what is wrong.

★★★ 2.20 A student, in treating the charge balance of a $\text{Ca}_3(\text{PO}_4)_2$ aqueous solution, wrote $[\text{Ca}^{2+}] = [\text{PO}_4^{3-}]$. Write the correct charge balance equation and explain why his writing is wrong.

★★★ 2.21 A student determined Al^{3+} by the back titration method and directly used the volume of the Zn^{2+} standard solution consumed in the back-titration to calculate the content of Al^{3+} : $n(\text{Al}^{3+}) = c(\text{Zn}^{2+}) \times V(\text{Zn}^{2+})$. Is this correct? If not, what is the correct formula?

Answer

2.1 (Omitted)

2.2 (Omitted)

2.3 1. positive error 2. cannot be determined 3. cannot be determined 4. negative error

2.4 (Omitted)

2.5 (Omitted)

2.6 0.750 mol/L, 0.02000 mol/L, 0.0400 mol/L and 0.200 mol/L, 0.0365 mol/L.

2.7 3.028%

2.8 0.1000 mol/L

2.9 0.1002 mol/L

2.10 0.01001 mol/L

2.11 0.2881 g

2.12 (Omitted)

2.13 (Omitted)

2.14 (Omitted)

2.15 11.46%

2.16 (Omitted)

2.17 (Omitted)

2.18 (Omitted)

2.19 (Omitted)

2.20 (Omitted)

2.21 (Omitted)

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